# Chemistry

# Class 11 (English Medium)

# Solved Exercises — Sindh Textbook Board

12 chapters — 267 questions

# Chapter 1: Stoichiometry

## Multiple Choice Questions: 1. Choose the correct answer

(i) If the volume occupied by oxygen gas (O2) at STP is 44.8dm3, the number of molecules of O2 in the vessels are:

(a) 3.01 x 10^23

(b) 6.02 x 10^23

✓ (c) 12.04 x 10^23

(d) 24.08 x 10^23

Reason: 44.8 dm3 at STP = 44.8/22.4 = 2 moles, and 2 x 6.02 x 10^23 = 12.04 x 10^23 molecules.

(ii) The number of carbon atoms in 1 mole of sugar (C12H22O11) are approximately:

(a) 6 x 10^23

(b) 24 x 10^23

(c) 60 x 10^23

✓ (d) 72 x 10^23

Reason: Each molecule has 12 carbon atoms, so 1 mole contains 12 x 6.02 x 10^23 = about 72 x 10^23 carbon atoms.

(iii) In the reaction 2Na + 2H2O -> 2NaOH + H2, if 23g of Na reacts with excess of water, the volume of hydrogen gas (H2) liberated at STP should be:

✓ (a) 11.2dm3

(b) 22.4dm3

(c) 33.6dm3

(d) 44.8dm3

Reason: 23g Na = 1 mole; 2 moles Na give 1 mole H2, so 1 mole Na gives 0.5 mole H2 = 0.5 x 22.4 = 11.2 dm3 at STP.

(iv) Which of the following sample of substances contains the same number of atoms as that of 20g calcium:

(a) 16g S

(b) 20g C

(c) 19g K

(d) 24g Mg

Reason: 20g Ca = 20/40 = 0.5 mole of atoms, and 16g S = 16/32 = 0.5 mole of atoms, so both contain the same number of atoms.

Question: (v) Which of the following statement is incorrect? (a) The mass of 1 mole Cl2 gas is 35.5g (b) One mole of H2 gas contains 6.02 x 10^23 molecules of H2 (c) Number of atoms in 23g Na and 24g Mg are equal (d) One moles of O2 at S.T.P occupy 22.4 dm3 volume

Answer: (a) — The molar mass of Cl2 is 2 x 35.5 = 71g, not 35.5g, so statement (a) is incorrect; all other statements are correct.

Question: (vi) For Avogadro's number, this statement is incorrect: (a) It is the no. of particles in one moles of any substances (b) Its numerical value is 6.02 x 10^23 (c) Its value change if temperature increases (d) Its value change if number of moles increases

Answer: (c) — Avogadro's number is a constant (6.02 x 10^23 per mole); it does not change with temperature, so statement (c) is incorrect.

Question: (vii) The minimum number of moles are present in: (a) 1 dm3 of methane gas at STP (b) 5 dm3 of helium gas at STP (c) 10 dm3 of hydrogen gas at STP (d) 22.4 dm3 of chlorine gas at STP

Answer: (a) — Moles = volume/22.4: (a) 0.045, (b) 0.22, (c) 0.45, (d) 1.0

mole, so 1 dm3 of methane has the minimum number of moles.

(viii) Number of atoms in 60g carbon are:

(a) 3.01 x 10^23

✓ (b) 3.01 x 10^24

(c) 6.02 x 10^23

(d) 6.02 x 10^24

Reason: 60g carbon = 60/12 = 5 moles, and 5 x 6.02 x 10^23 = 3.01 x 10^24 atoms.

(ix) Maximum number of molecules present in the following sample of gas:

(a) 100g O2

✓ (b) 100g CH4

(c) 100g CO2

(d) 100g Cl2

Reason: CH4 has the smallest molar mass (16 g/mol), so 100g CH4 = 6.25 moles, which is the largest amount and hence has the maximum number of molecules.

(x) Generally actual yield is:

(a) Greater than theoretical yield

✓ (b) Less than theoretical yield

(c) Equal to the theoretical yield

(d) Some times greater and some times less than theoretical yield

Reason: Due to side reactions, incomplete reaction and handling losses, the actual (practical) yield is generally less than the theoretical yield.

## Short Questions

Question: 1. Define the following: (i) Stoichiometry (ii) Exponential

Notation (iii) Molar Volume

Answer: (i) Stoichiometry is the study of the quantitative relationship between reactants and products in a chemical reaction using a balanced chemical equation. (ii) Exponential notation is a shorthand mathematical form in which a number is written as a coefficient multiplied by 10 raised to some power, e.g. 3652 = 3.652 x 10^3. (iii) Molar volume is the volume occupied by one mole of any gas at STP (273K and 1 atm pressure); for all ideal gases its value is 22.4 dm3.

Question: 2. Express the following numbers in exponential notation: (i) 3652 (ii) 0.0231 (iii) 0.000072

Answer: (i) 3652 = 3.652 x 10^3. (ii) 0.0231 = 2.31 x 10^-2. (iii) 0.000072 = 7.2 x 10^-5. In each case the decimal point is moved so that only one non-zero digit is left before it, and the power of 10 counts the places moved.

Question: 3. Express the following in simple numbers. (i) 3.26 x 10^-3 (ii) 1.921 x 10^2 (iii) 1.02 x 10^5

Answer: (i) 3.26 x 10^-3 = 0.00326. (ii) 1.921 x 10^2 = 192.1. (iii) 1.02 x 10^5 = 102000. A negative power moves the decimal point to the left and a positive power moves it to the right.

Question: 4. Define rounding of data. Give various rules of rounding of data.

Answer: Rounding of data is a method of reducing the figures in a given numerical value; it means reducing a number to the desired significant figures and adjusting the last reported digit. If the digit to be dropped is less than 5, the last retained digit stays unchanged. If it is greater than 5, the last retained digit is increased by one. If it is exactly 5, the last

retained digit is increased by one only if it is odd, and left unchanged if it is even.

## Descriptive Questions

**Question: 1. Define theoretical yield, actual yield and percent yield. Why the practical yield is often less than theoretical yield?**

Answer: Theoretical yield is the maximum amount of product calculated from a balanced chemical equation, assuming complete conversion of the limiting reactant. Actual or practical yield is the amount of product actually obtained during the experiment. Percent yield is the ratio of practical yield to theoretical yield multiplied by 100. Practical yield is often less because the reaction may not go to completion, side reactions may consume reactants, some product is lost during separation and purification, and the reaction may be reversible.

**Question: 2. What is meant by mole and Avogadro's number? How are they inter related to each other?**

Answer: A mole is the gram atomic mass, gram molecular mass or gram formula mass of a substance, and it contains 6.02 x 10^23 particles (atoms, molecules or ions). Avogadro's number is this number of particles present in one mole of any substance, denoted by NA. The two are interrelated because the mole is defined through Avogadro's number: one mole of every substance, whatever its mass in grams, always contains Avogadro's number of particles. Thus number of moles = number of particles / 6.02 x 10^23.

**Question: 3. What is meant by Avogadro's number? Explain concept of mole with the help of Avogadro's number.**

Answer: Avogadro's number is the number of particles present in one mole

of any substance, equal to 6.02 x 10^23. With its help, the mole becomes a counting unit like a dozen: just as one dozen means 12 objects, one mole means 6.02 x 10^23 particles. For example, one mole of carbon atoms weighs 12g and contains 6.02 x 10^23 atoms, while one mole of water weighs 18g and contains 6.02 x 10^23 molecules. This lets us convert between mass, moles and number of particles in calculations.

Question: 4. What is a Limiting reactant? How does it control the amount of product formed in a chemical reaction.

Answer: The limiting reactant is the reactant which is consumed entirely when a chemical reaction is completed, while the other reactant is left partly unused (reactant in excess). It controls the amount of product because the reaction stops as soon as the limiting reactant is used up, even though the excess reactant is still present. Therefore the quantity of product is always calculated from the amount of the limiting reactant. For example, if hydrogen is limiting in the burning of hydrogen in oxygen, the water formed depends only on the hydrogen available.

## Numerical Questions

Question: 1. Calculate the number of moles and molecules in: (i) 38g of carbon disulphide (CS2) (ii) 68.4g of sucrose (C12H22O11)

Answer: (i) Molar mass of CS2 = 12 + 2(32) = 76 g/mol. Moles = 38/76 = 0.499 moles (about 0.5 mole). Molecules = 0.499 x 6.02 x 10^23 = 3.0 x 10^23 molecules. (ii) Molar mass of sucrose = 12(12) + 22(1) + 11(16) = 342 g/mol. Moles = 68.4/342 = 0.199 moles (about 0.2 mole). Molecules = 0.199 x 6.02 x 10^23 = 1.19 x 10^23 molecules.

Question: 2. Ammonia gas can be produced by heating together the solid NH4Cl and Ca(OH)2. 2NH4Cl + Ca(OH)2 -> 2NH3 + CaCl2 + 2H2O. If a

mixture containing 100g of each of these solids is heated, determine the limiting reactant and the mass of NH3 gas produced.

Answer: Moles of NH4Cl = 100/53.5 = 1.87 mol; moles of Ca(OH)2 = 100/74 = 1.35 mol. The equation needs 2 moles NH4Cl for 1 mole Ca(OH)2, so 1.87 mol NH4Cl needs only 0.935 mol Ca(OH)2, which is available in excess. Hence NH4Cl is the limiting reactant. NH3 formed = 1.87 mol, and mass of NH3 = 1.87 x 17 = 31.7g.

Question: 3. Aluminum chloride is used in the manufacturing of rubber. It is produced by allowing Aluminum to react with Cl2 gas at 650 degree C. 2Al(s) + 3Cl2(g) -> 2AlCl3(l). When 160g Aluminum reacts with excess of chlorine, 650g of AlCl3 is produced. What is the percentage yield of AlCl3?

Answer: Moles of Al = 160/27 = 5.93 mol, so theoretical moles of AlCl3 = 5.93 mol. Theoretical mass = 5.93 x 133.5 = about 791g. Percent yield = (actual yield / theoretical yield) x 100 = (650/791) x 100 = 82.17%.

Question: 4. 1.6g of a sample of gas occupies a volume of 1.12 dm3 at STP. Determine the molar mass of the substance.

Answer: At STP one mole of a gas occupies 22.4 dm3. Moles of gas = 1.12/22.4 = 0.05 mol. Molar mass = mass / moles = 1.6/0.05 = 32 g/mol.

Question: 5. Silver sulphide (Ag2S) is an anti microbial agent. In an experiment 24.8g Ag2S is reacted with the excess of hydrochloric acid as given in the following reaction. Ag2S(s) + 2HCl(aq) -> 2AgCl(s) + H2S(g). Calculate the (i) Mass of AgCl formed (ii) Volume of H2S produced at STP (At. Mass of Ag is 108 g/mol and S is 32 g/mol)

Answer: Molar mass of Ag2S = 2(108) + 32 = 248 g/mol, so moles = 24.8/248 = 0.1 mol. (i) AgCl formed = 2 x 0.1 = 0.2 mol; molar mass of AgCl = 108 + 35.5 = 143.5 g/mol, so mass = 0.2 x 143.5 = 28.7g. (ii) H2S

formed = 0.1 mol, so volume at STP = 0.1 x 22.4 = 2.24 dm3.

Question: 6. Calculate each of the following quantities. (i) Number of moles in 6.4g of SO2 (ii) Mass in gram of 4.5 moles of ethyne (C2H2) (iv) Volume in cm3 of 38.4g O2 gas at STP (v) Number of molecules of 126g water (vi) Mass in gram of 4.8 x 10^24 atoms of sodium (vi) Number of formula units in 333g of CaCl2

Answer: (i) Moles = 6.4/64 = 0.1 mol. (ii) Molar mass of C2H2 = 26 g/mol, mass = 4.5 x 26 = 117g. (iv) Moles of O2 = 38.4/32 = 1.2 mol; volume = 1.2 x 22400 = 26880 cm3 at STP. (v) Moles of water = 126/18 = 7 mol; molecules = 7 x 6.02 x 10^23 = 42.14 x 10^23. (vi) Moles of Na = 4.8 x 10^24 / 6.02 x 10^23 = 7.97 mol; mass = 7.97 x 23 = 183.39g. (vi) Moles of CaCl2 = 333/111 = 3 mol; formula units = 3 x 6.02 x 10^23 = 18.06 x 10^23.

# Chapter 2: Atomic Structure

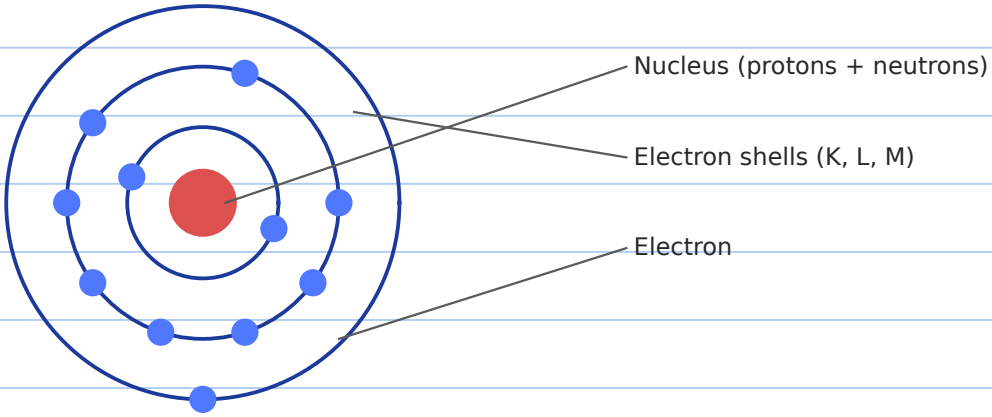



Fig: Bohr's Model of the Atom (Sodium, 2,8,1)

## Multiple Choice Questions: 1. Choose the correct answer

(i) Bohr's theory is not applicable to which of the following species.

(a) H

✓ (b) H-

(c) He+1

(d) Li+2

Reason: Bohr's theory applies only to one-electron species like H, He+ and Li+2; H- has two electrons, so the theory is not applicable to it.

(ii) Nitrogen has the electronic configuration 1s2 2s2 2p1x 2p1y 2p1z and not 1s2 2s2 2p2x 2p1y. This is determined by.

(a) Aufbau principle

(b) Pauli's principle

✓ (c) Hund's rule

(d) n + l rule

Reason: Hund's rule says electrons fill degenerate orbitals singly with the same spin before pairing, so the three p electrons of nitrogen

occupy separate p orbitals.

(iii) Quantum number values for 3s orbital are

(a) n = 0, l = 1

(b) n = 1, l = 0

(c) n = 3, l = 1

✓ (d) n = 3, l = 0

Reason: For a 3s orbital the principal quantum number n = 3, and for an s orbital the azimuthal quantum number l = 0.

(iv) The radius of first orbit of hydrogen atom is .....

(a) 529 A

(b) 52.9 A

(c) 5.29 A

✓ (d) 0.529 A

Reason: The Bohr radius of the first orbit of hydrogen is 0.529 A, given by r = 0.529 x n^2 / Z with n = 1 and Z = 1.

(v) Line spectrum is used as a tool for the identification of ....

(a) Colors

(b) Electrons

✓ (c) Elements

(d) Molecules

Reason: Each element gives a unique line spectrum, so the spectrum acts as a fingerprint for the identification of elements.

(vi) In 1935 A.D. James Chadwick was awarded Nobel Prize because ...

(a) He discovered proton

✓ (b) He discovered neutron

(c) He determined the radius of hydrogen atom

(d) He gave the rules for electronic configuration

Reason: Chadwick discovered the neutron in 1932 and was awarded the Nobel Prize in 1935 for this discovery.

(vii) When 4d orbital is filled, the next electron enter into

(a) 5s

✓ (b) 5p

(c) 5d

(d) 6s

Reason: According to the (n + l) rule the order of filling after 4d is 5p, because 5p has a lower (n + l) value than 5d and 6s at that stage.

(viii) Which of the following is not an iso electronic pair

(a) Na+ and Ne

(b) Na+ and F-

✓ (c) Na and Ca

(d) Na+ and Mg+2

Reason: Na has 11 electrons and Ca has 20 electrons, so they are not isoelectronic; the other pairs each have the same number of electrons (10).

(ix) Balmer series appears in the hydrogen spectrum if electron jump from any appropriate higher energy orbit to

✓ (a) Second orbit

(b) Third orbit

(c) Fourth orbit

(d) Fifth orbit

Reason: The Balmer series is produced when electrons fall from higher orbits (n = 3, 4, 5 ...) to the second orbit (n = 1 level is 2), and it lies

in the visible region.

(x) The shape of orbital for which l = 0 is

(a) Spherical

(b) Dumbbell

(c) Double dumbbell

(d) Complicated

Reason: l = 0 corresponds to an s orbital, which is spherical in shape around the nucleus.

## Short Questions

Question: 1. Differentiate between Continuous and Line spectrum.

Answer: A continuous spectrum is produced when white light passes through a prism; it consists of a band of seven colours (VIBGYOR) which diffuse into each other with no distinct boundary line. A line spectrum consists of separate bright or dark lines with empty spaces between them, and each line has a definite wavelength. A continuous spectrum is given by hot solids and liquids, while a line spectrum is given by gases and vapours in a discharge tube. Line spectra are characteristic of elements and are used for their identification.

Question: 2. Give three properties of each alpha, beta and gamma rays.

Answer: Alpha rays consist of positively charged particles (helium nuclei); they deflect towards the negative pole in an electric field, have low penetrating power and high ionizing power. Beta rays consist of negatively charged particles (electrons); they deflect towards the positive pole, have greater penetrating power than alpha rays and moderate ionizing power. Gamma rays are electromagnetic radiations of very short wavelength with no charge; they pass through electric and magnetic fields without

deflection, have the highest penetrating power and low ionizing power.

Question: 3. What is the shape of orbitals for which l = 0 and l = 1.

Answer: For l = 0 the orbital is an s orbital, which is spherical in shape, with the electron density distributed equally in all directions around the nucleus. For l = 1 the orbital is a p orbital, which is dumbbell shaped, having two lobes on opposite sides of the nucleus separated by a nodal plane. There are three p orbitals (px, py, pz) oriented along the three axes. The size of both types of orbital increases with the principal quantum number n.

Question: 4. How does an orbital different from orbit?

Answer: An orbit is a well-defined circular path around the nucleus in which an electron revolves, as proposed by Bohr; its position and momentum can be known at the same time. An orbital is a three-dimensional region of space around the nucleus where the probability of finding an electron is maximum (about 95%). An orbit is a two-dimensional concept with a fixed radius, whereas an orbital has a definite shape (spherical, dumbbell, etc.) but no fixed path. The orbital concept is based on quantum mechanics and the uncertainty principle.

Question: 5. Explain why the filling of electron in 4s orbital takes place prior to 3d?

Answer: The filling order is decided by the (n + l) rule: the orbital with the lower (n + l) value is filled first. For 4s, (n + l) = 4 + 0 = 4, while for 3d, (n + l) = 3 + 2 = 5. Since 4s has the lower (n + l) value, it has lower energy and is filled before 3d. This is why potassium and calcium place their electrons in 4s before 3d begins to fill in the transition metals.

Question: 6. Mention the defects of Bohr's atomic model.

Answer: Bohr's model does not explain the spectra of atoms having more than one electron, such as helium. It cannot explain the fine structure of spectral lines observed under high resolution, nor the splitting of lines in electric and magnetic fields (Stark and Zeeman effects). It treats the electron only as a particle and ignores its wave nature, and it violates Heisenberg's uncertainty principle by fixing both the position and momentum of the electron. It also gives no satisfactory explanation of chemical bonding and molecular formation.

Question: 7. Write down the electronic configuration of the following. (i) Fe (Z = 26), (ii) Br- (Z = 35), (iii) Ca+2 (Z = 20)

Answer: (i) Fe (26): 1s2 2s2 2p6 3s2 3p6 4s2 3d6. (ii) Br- has 35 + 1 = 36 electrons: 1s2 2s2 2p6 3s2 3p6 4s2 3d10 4p6, the same as krypton. (iii) Ca+2 has 20 - 2 = 18 electrons: 1s2 2s2 2p6 3s2 3p6, the same as argon. In each case electrons are filled in order of increasing energy following the Aufbau principle.

## Descriptive Questions

Question: 1. (a) State the postulates of Bohr atomic theory. (b) Drive an expression for the frequency of radiation emitted from an electron. Given that E = -me^4 / 8e0^2 h^2 n^2

Answer: (a) Bohr postulated that electrons revolve around the nucleus in fixed circular orbits of definite energy without radiating energy. The angular momentum of the electron is an integral multiple of h/2pi. Energy is absorbed when an electron jumps to a higher orbit and emitted when it falls to a lower orbit, with frequency given by E2 - E1 = hv. (b) The energy of the electron in orbit n is En = -me^4/(8e0^2 h^2 n^2). For a jump from n2 to n1, the energy emitted is DE = E(n2) - E(n1) = (me^4/8e0^2 h^2)(1/n1^2 - 1/n2^2). Since DE = hv, the frequency is v =

(me^4/8e0^2 h^3)(1/n1^2 - 1/n2^2), which explains the lines of the hydrogen spectrum.

Question: 2. What are X-rays? How are they produced? Give their properties and uses.

Answer: X-rays are short wavelength, highly penetrating electromagnetic radiations. They are produced when fast-moving electrons (cathode rays) strike a metal target of high atomic mass, such as tungsten, in a discharge tube; part of their kinetic energy changes into X-rays. Their properties are: they travel in straight lines at the speed of light, are not deflected by electric or magnetic fields, penetrate flesh but are stopped by bones and heavy metals, and affect photographic plates. They are used in medical diagnosis and surgery, in industry for detecting defects in materials, in the study of crystal structure, and in the treatment of some diseases.

Question: 3. State and illustrate the following rules of electronic configuration. a) Pauli's exclusion rule b) Hund's rule of maximum multiplicity

Answer: (a) Pauli's exclusion principle states that in an atom no two electrons can have the same set of four quantum numbers; at least one quantum number must be different. Hence an orbital can hold at most two electrons, and they must have opposite spins, illustrated as one up arrow and one down arrow in the same box. (b) Hund's rule states that when degenerate orbitals are available, electrons fill them singly with the same spin direction so as to give the maximum number of unpaired electrons. For example, the three p orbitals of nitrogen each receive one electron before any pairing begins.

Question: 4. Explain hydrogen spectrum in term of Bohr's theory.

Answer: According to Bohr's theory, in the ground state the electron of hydrogen stays in the first orbit. When energy is supplied, it jumps to higher orbits (n = 2, 3, 4 ...). When it falls back, it emits radiation of fixed frequency, producing a line spectrum. Falls to n = 1 give the Lyman series in the ultraviolet region, falls to n = 2 give the Balmer series in the visible region, and falls to n = 3, 4 and 5 give the Paschen, Brackett and Pfund series in the infrared region. Each line corresponds to a definite energy difference between two orbits.

Question: 5. Describe four quantum numbers needed to specify an electron in an atom. Write all possible values of l, m and s for n = 2.

Answer: The principal quantum number n describes the size and energy of the shell and takes values 1, 2, 3 .... The azimuthal quantum number l describes the shape of the orbital and ranges from 0 to (n - 1). The magnetic quantum number m describes the orientation of the orbital in space and ranges from -l to +l. The spin quantum number s describes the spin of the electron and is +1/2 or -1/2. For n = 2: l = 0, 1; for l = 0, m = 0; for l = 1, m = -1, 0, +1; and for each electron s = +1/2 or -1/2.

## Numerical Questions

Question: 1. Calculate the radius of hydrogen in 3rd orbit (Bohr constant for hydrogen is 0.529A).

Answer: Bohr's formula for the radius is rn = 0.529 x n^2 / Z (in A). For hydrogen, Z = 1 and for the 3rd orbit n = 3. So r3 = 0.529 x 3^2 / 1 = 0.529 x 9 = 4.761 A. Hence the radius of the third orbit of hydrogen is about 4.76 A.

Question: 2. A photon of wave number 23 x 10^5 m^-1 is emitted when electron undergoes a transition from a higher energy orbit to n = 2.

Determine the orbit from which electron fall and also the spectral line appears in this transition of electron. (The value of Rydberg constant is 1.09678 x 10^7 m^-1).

Answer: Use the Rydberg relation: wave number = RH (1/n1^2 - 1/n2^2), with n1 = 2. So 23 x 10^5 = 1.09678 x 10^7 (1/4 - 1/n2^2), giving (1/4 - 1/n2^2) = 0.2097. Then 1/n2^2 = 0.25 - 0.2097 = 0.0403, so n2^2 = 24.8 and n2 = about 5. The electron therefore falls from the 5th orbit to the 2nd orbit. Since the lower orbit is n = 2, the line belongs to the Balmer series and appears in the visible region.

# Chapter 3: Theories of Covalent Bonding and Shapes of Molecules

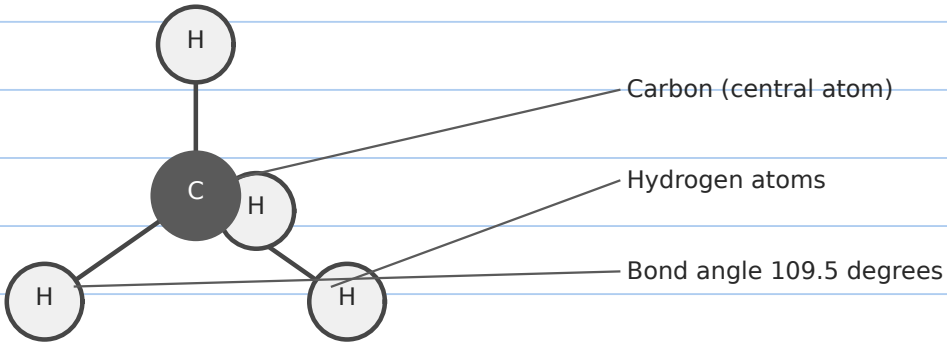



Fig: Tetrahedral Shape of Methane (CH4)

## Multiple Choice Questions: 1. Choose the correct answer

(i) If the bond angle in AB2 type molecule is 104.5 degree, it geometry should be:

(a) Linear

(b) Pyramidal

✓ (c) Bent

(d) Planar Trigonal

Reason: An AB2 molecule with a bond angle of 104.5 degree (like H2O) has two bond pairs and two lone pairs, giving it a bent (angular) geometry.

(ii) The highest bond energy in the following is:

(a) Cl - Cl

✓ (b) H - F

(c) H - O

(d) H - N

Reason: The H-F bond is very short and strongly polar due to the small size and high electronegativity of fluorine, so it has the highest bond energy among the given bonds.

(iii) The molecule which has zero dipole moment is:

(a) NH3

(b) HCl

(c) H2O

✓ (d) CCl4

Reason: CCl4 is a regular tetrahedron; the four equal C-Cl bond dipoles cancel each other, so the molecule has zero dipole moment.

(iv) The molecule which has maximum bond angle:

✓ (a) CS2

(b) H2O

(c) NH3

(d) BF3

Reason: CS2 is a linear molecule with a bond angle of 180 degree, which is greater than BF3 (120 degree), NH3 (107 degree) and H2O (104.5 degree).

(v) The shape and hybridization of BCl3 molecule is:

(a) Tetrahedral and sp3

(b) Linear and sp

✓ (c) Planar trigonal and sp2

(d) Angular and sp3

Reason: Boron in BCl3 has three bond pairs and no lone pair, so it undergoes sp2 hybridization and the molecule is planar trigonal with

120 degree angles.

(vi) Amongst the following molecules which one has trigonal pyramidal shape?

(a) SO2

(b) CO2

✓ (c) NH3

(d) C2H4

Reason: NH3 has three bond pairs and one lone pair on nitrogen; the lone pair pushes the bonds down, giving a trigonal pyramidal shape.

(vii) A simple covalent molecule possesses two bond pairs and two lone pairs around the central atom, its shape should be:

(a) Linear

(b) Planar trigonal

✓ (c) Angular

(d) Tetrahedral

Reason: Two bond pairs and two lone pairs give an angular (bent) shape, as in H2O, because the lone pairs repel strongly and reduce the bond angle.

(viii) The correct relation between Debye and coulomb meter is:

✓ (a) 1D = 3.33 x 10^-30 Cm

(b) 1D = 1.6 x 10^-19 Cm

(c) 1D = 1.88 x 10^-42 Cm

(d) 1D = 1.23 x 10^-8 Cm

Reason: The Debye is the practical unit of dipole moment, and 1 Debye = 3.33 x 10^-30 coulomb meter (Cm).

(ix) The bond order of N2 molecule is:

(a) 0

(b) 1

(c) 2

(d) 3

Reason: In N2, the number of bonding electrons is 10 and antibonding electrons is 4, so bond order = (10 - 4)/2 = 3, corresponding to a triple bond.

(x) The number of sigma and pi bonds in C2H4 molecule are respectively:

(a) 3 and 1

(b) 2 and 2

(c) 5 and 1

(d) 4 and 2

Reason: Ethene has four C-H single bonds and one C-C double bond; the single bonds and one part of the double bond give 5 sigma bonds, and the remaining part of the double bond is 1 pi bond.

## Short Questions

Question: 1. Define the following: (i) Bond Energy (ii) Bond length

Answer: (i) Bond energy is the amount of energy required to break one mole of a particular type of bond in a molecule; it is measured in KJ/mol and shows the strength of the bond. (ii) Bond length is the distance between the nuclei of two bonded atoms in a molecule, measured in Angstrom (A). It depends on the size of the atoms and the bond order: a higher bond order means a shorter bond length. For example, a C-C single bond is longer than a C=C double bond.

Question: 2. The Dipole moment of water is 1.85D but CO2 has zero dipole moment why?

Answer: Water has a bent shape, so the dipoles of the two O-H bonds do not cancel; they add up to give a resultant dipole moment of 1.85D, making water polar. Carbon dioxide is a linear molecule (O=C=O), in which the two equal C=O bond dipoles point in exactly opposite directions. Hence they cancel each other completely and the net dipole moment of CO2 is zero. This shows that dipole moment depends not only on bond polarity but also on molecular geometry.

Question: 3. Differentiate between the following: (i) VBT and MOT (ii) Sigma bond and pi bond (iii) Bonding M.O and Anti bonding M.O

Answer: (i) In valence bond theory (VBT) the covalent bond is formed by overlapping of atomic orbitals and the shared pair stays localized between the two atoms; in molecular orbital theory (MOT) atomic orbitals combine to form molecular orbitals spread over the whole molecule. (ii) A sigma bond is formed by head-to-head overlapping along the internuclear axis and is strong; a pi bond is formed by side-by-side overlapping of p orbitals and is weaker. (iii) A bonding molecular orbital has lower energy than the atomic orbitals and holds the atoms together, while an antibonding molecular orbital has higher energy and tends to separate the atoms.

Question: 4. Explain the following with scientific reasons. (i) Sigma bond is stronger than pi bond. (ii) HF has greater ionic character than HCl. (iii) Bond energy of molecules possessing multiple bonds is high.

Answer: (i) In a sigma bond the overlapping of orbitals is head-on and extensive, while in a pi bond the side-by-side overlapping is partial and less effective, so a sigma bond is stronger. (ii) Fluorine is much more electronegative than chlorine, so the electronegativity difference between H and F is larger than between H and Cl; hence the H-F bond is more polar and has greater ionic character. (iii) A multiple bond (double or triple)

contains one sigma and one or two pi bonds, and more energy is needed to break all of them together; also the atoms are held closer, so the bond energy is high.

Question: 5. Oil is insoluble in water but soluble in hexane explain why?
Answer: Oil is a non-polar covalent substance, while water is a polar solvent. According to the rule "like dissolves like", the polar water molecules attract each other strongly through hydrogen bonding and do not allow the non-polar oil molecules to mix, so oil forms a separate layer. Hexane is also non-polar, and the weak London forces between oil and hexane molecules are comparable with the forces within each liquid. Therefore oil dissolves in hexane, just as non-polar substances dissolve in non-polar solvents.

Question: 6. Which hybridization would you expect for the central atom in each of the following? $BeCl_2$, $BF_3$, $CH_4$, $C_2H_2$
Answer: In $BeCl_2$, beryllium has two bond pairs, so it is sp hybridized and the molecule is linear. In $BF_3$, boron has three bond pairs, so it is $sp^2$ hybridized and the molecule is planar trigonal. In $CH_4$, carbon has four bond pairs, so it is $sp^3$ hybridized and the molecule is tetrahedral. In $C_2H_2$ (ethyne), each carbon forms two sigma bonds (one with H and one with C), so each carbon is sp hybridized and the molecule is linear.

## Descriptive Questions

Question: 1. (a) Write down the postulate of VSEPR theory (b) Predict the shape of following molecules on the basis of VSEPR theory. $AlCl_3$, $CCl_4$, $PH_3$, $H_2S$
Answer: (a) VSEPR theory postulates that the shape of a molecule is determined by the repulsion between electron pairs around the central

atom; the pairs arrange themselves as far apart as possible to minimize repulsion. Repulsion decreases in the order lone pair-lone pair > lone pair-bond pair > bond pair-bond pair. (b) AlCl3 has 3 bond pairs and no lone pair, so it is planar trigonal. CCl4 has 4 bond pairs and no lone pair, so it is tetrahedral. PH3 has 3 bond pairs and 1 lone pair, so it is trigonal pyramidal. H2S has 2 bond pairs and 2 lone pairs, so it is angular (bent).

Question: 2. What do you mean by Hybridization? Explain sp3 hybridization in CH4 molecule and sp2 hybridization in C2H4 molecule.
Answer: Hybridization is the mixing of atomic orbitals of nearly equal energy to produce an equal number of new orbitals of the same energy, called hybrid orbitals. In CH4, the 2s and three 2p orbitals of carbon mix to form four sp3 hybrid orbitals directed towards the corners of a tetrahedron; each overlaps with the 1s orbital of hydrogen, giving a tetrahedral molecule with 109.5 degree angles. In C2H4, each carbon mixes its 2s and two 2p orbitals to form three sp2 hybrid orbitals in one plane at 120 degree, used for sigma bonds with two hydrogens and the other carbon. The unhybridized p orbital on each carbon overlaps sideways to form the pi bond of the C=C double bond.

Question: 3. What are the basic postulates of valence bond theory? Explain various type of overlapping lead to sigma and pi bond.
Answer: Valence bond theory postulates that a covalent bond is formed by the overlapping of half-filled atomic orbitals of two atoms, and the shared electron pair is localized between the two nuclei. The overlapping takes place in the direction of maximum electron density, the atoms keep their identity in the molecule, and greater overlapping gives a stronger bond. End-to-end (head-on) overlapping along the internuclear axis, such as s-s, s-p and p-p axial overlap, produces a sigma bond which is strong and

allows free rotation. Side-by-side overlapping of parallel p orbitals above and below the axis produces a pi bond, which is weaker and restricts rotation about the bond.

Question: 4. Draw a molecular orbital diagram of O2 molecule and (i) Write down MO electronic configuration (ii) Determine bond order (iii) Explain the paramagnetic behavior

Answer: The molecular orbital diagram of O2 is obtained by combining the atomic orbitals of two oxygen atoms; the 16 electrons fill the molecular orbitals in order of increasing energy. (i) MO configuration: (sigma1s)2 (sigma*1s)2 (sigma2s)2 (sigma*2s)2 (sigma2pz)2 (pi2px)2 (pi2py)2 (pi*2px)1 (pi*2py)1. (ii) Bond order = (bonding electrons - antibonding electrons)/2 = (10 - 6)/2 = 2, so O2 has a double bond. (iii) The two antibonding pi* orbitals each contain one unpaired electron. Because of these two unpaired electrons, O2 is paramagnetic, and this behaviour can only be explained by molecular orbital theory, not by valence bond theory. (As instructed, no diagram is drawn here; the diagram places the two pi* orbitals at the top with one unpaired electron each.)

# Chapter 4: State of Matter I: Gas

## Multiple Choice Questions: 1. Choose the correct answer

(i) According to Graham's Law of diffusion, the ratio of diffusion of H2 and O2 are respectively:

(a) 1:2

(b) 2:1

(c) 1:4

✓ (d) 4:1

Reason: Rate of diffusion is inversely proportional to the square root of molar mass; sqrt(32/2) = 4, so H2 diffuses 4 times faster than O2, a ratio of 4:1.

(ii) Collection of gas over water is an example of:

(a) Graham's law

✓ (b) Dalton's law

(c) Avogadro's law

(d) Gaylussac law

Reason: When a gas is collected over water, the total pressure is the sum of the partial pressure of the gas and the vapour pressure of water, which is Dalton's law of partial pressure.

(iii) The molar volume of oxygen gas is maximum at:

(a) 0 degree C and 1 atm

(b) 0 degree C and 2 atm

✓ (c) 25 degree C and 1 atm

(d) 25 degree C and 2 atm

Reason: Molar volume increases with temperature and decreases with

pressure, so it is maximum at the highest temperature and lowest pressure given, 25 degree C and 1 atm.

(iv) The diffusion rate of C3H8 and CO2 are same because:

(a) Both are poly atomic gases

(b) Both are denser than air

✓ (c) Both have same molar mass

(d) Both contains carbon atoms

Reason: C3H8 and CO2 both have molar mass 44 g/mol, and by Graham's law gases of equal molar mass diffuse at the same rate.

(v) The volume of gas would be theoretically zero at:

(a) 0 degree C

✓ (b) 0 K

(c) 273 K

(d) 273 degree C

Reason: By Charles's law the volume of a gas decreases with temperature and becomes theoretically zero at absolute zero, 0 K (-273 degree C).

(vi) Real gas reaches the ideal behavior at:

(a) Low temperature and low pressure

(b) High temperature and high pressure

(c) Low temperature and high pressure

✓ (d) High temperature and low pressure

Reason: At high temperature and low pressure the gas molecules are far apart and intermolecular forces become negligible, so a real gas behaves almost ideally.

(vii) Which one of the following statement is incorrect about the gas

molecules?

(a) They have large spaces

(b) They possess kinetic energy

(c) Their collision is elastic

✓ (d) Their molar mass depends upon temperature

Reason: Molar mass is a fixed property of a gas and does not depend on temperature; the other statements about gas molecules are correct.

(viii) If the Kelvin temperature of ideal gas is increase to double and pressure is reduce to one half, the volume of gas will:

(a) Remains same

(b) Double

(c) Reduced to half

✓ (d) Four time

Reason: From V proportional to T/P: doubling T doubles V and halving P doubles it again, so the volume becomes four times.

(ix) The molar volume of oxygen gas is 22.4 dm3 at:

✓ (a) 0 degree C and 1 atm

(b) 25 degree C and 0.5 atm

(c) 0 K and 1 atm

(d) 25 K and 0.5 atm

Reason: The molar volume of any ideal gas is 22.4 dm3 at STP, that is 0 degree C (273K) and 1 atm pressure.

Question: (x) Under similar condition CH4 gas diffuses..... times faster than SO2 gas: (a) 1.5 time (b) 2 times (c) 4 times (d) 16 times

Answer: (c) — As given in the book's answer key. (Note: Graham's law

gives sqrt(64/16) = 2, i.e. CH4 diffuses 2 times faster; the book key marks (c) 4 times.)

## Short Questions

**Question: 1. State the following gas laws. (i) Avogadro's law (ii) Charles law**

Answer: (i) Avogadro's law states that equal volumes of all gases contain an equal number of moles (or molecules) at the same temperature and pressure; the volume of a gas is directly proportional to its number of moles when temperature and pressure are constant, V proportional to n. (ii) Charles's law states that the volume of a given mass of a gas is directly proportional to its absolute temperature when pressure is kept constant, V proportional to T. It is used to define absolute zero, the temperature at which the volume of a gas would become zero.

**Question: 2. State main postulates of kinetic molecular theory of gas.**

Answer: The main postulates are: a gas consists of a large number of tiny particles (molecules) which are far apart, so the actual volume of the molecules is negligible compared with the volume of the gas. The molecules are in continuous rapid motion in all directions and collide with each other and with the walls of the container; these collisions are perfectly elastic. There are no forces of attraction between the molecules. The average kinetic energy of the molecules is directly proportional to the absolute temperature, and the pressure of the gas is produced by the collisions of molecules with the walls.

**Question: 3. Explain the following. (i) Pressure and its various units (ii) Absolute zero**

Answer: (i) Pressure is the force applied per unit area on the surface of an

object; for a gas it results from collisions of molecules with the walls of the container. Its SI unit is the pascal (N/m2), and it is also measured in atmosphere, torr and mm of Hg, where 1 atm = 760 torr = 760 mmHg = 101325 pascal. (ii) Absolute zero is the lowest possible temperature, 0 K or -273.15 degree C, at which the volume of an ideal gas would become zero. In practice all gases liquefy before reaching it, and at absolute zero the kinetic energy and motion of the particles are at their minimum.

Question: 4. What is liquid air? Mention its three uses.

Answer: Liquid air is ordinary air which, on compression and cooling to an extremely low temperature (about -190 degree C), becomes liquefied. It is a pale blue liquid mixture mainly of liquid nitrogen and liquid oxygen. Its uses are: it provides a very low temperature for scientific research and refrigeration; fractional distillation of liquid air gives nitrogen, oxygen and noble gases for industry and hospitals; and liquid oxygen obtained from it is used in rockets, welding and medical treatment. Liquid air must be stored in special insulated vessels because it evaporates very quickly.

Question: 5. What is plasma? Give its significance in daily life.

Answer: Plasma is the fourth state of matter, consisting of a mixture of ions, electrons and un-ionized atoms and molecules. It is formed when a gas is heated to a very high temperature or subjected to a strong electric field, so that its atoms become ionized. In daily life plasma is significant in fluorescent tubes and neon sign lights, in plasma television screens, and in plasma torches used for cutting metals. The sun and other stars consist of plasma, and it is also being studied for producing electricity by nuclear fusion.

## Descriptive Questions

Question: 1. State and explain Dalton's law of partial pressure. Give practical applications of Dalton's law.

Answer: Dalton's law states that the total pressure of a mixture of two or more non-reacting gases in a fixed volume enclosed container is equal to the sum of the partial pressures of the individual gases, P(total) = P1 + P2 + P3 + ... The partial pressure of a gas is the pressure it would exert if it occupied the vessel alone, and it can be calculated from its mole fraction multiplied by the total pressure. Its practical applications include collecting a gas over water and correcting its pressure for water vapour, calculating pressures in gas mixtures used in deep-sea diving, and working out the composition of air and respiratory gases. It is also used in industry wherever gases are stored and mixed in cylinders.

Question: 2. Derive general gas equation. Also deduce the value of R in atm dm3/mol.K and J/mol.K.

Answer: Boyle's law gives V proportional to 1/P at constant T, and Charles's law gives V proportional to T at constant P. Combining them, V proportional to T/P, or PV proportional to T. For n moles this becomes PV = nRT, the general (ideal) gas equation, where R is the universal gas constant. At STP one mole occupies 22.4 dm3 at 1 atm and 273K, so R = PV/nT = (1 x 22.4)/(1 x 273) = 0.0821 atm dm3/mol.K. In SI units, R = (101325 Pa x 0.0224 m3)/(1 x 273) = 8.314 J/mol.K.

Question: 3. How an ideal gas is differentiated from real gas? What are the causes of deviation of real gas from ideal behavior? Explain this deviation at low temperature and high pressure.

Answer: An ideal gas obeys the gas laws and the equation PV = nRT at all temperatures and pressures; its molecules are assumed to have no volume and no intermolecular attractions. A real gas obeys the van der

Waals equation and shows ideal behaviour only at high temperature and low pressure. The causes of deviation are the two faulty assumptions of the kinetic theory: real gas molecules do occupy a finite volume, and attractive forces do exist between them. At low temperature the molecules move slowly, so attractions pull them together and the volume and pressure are less than expected. At high pressure the molecules are crowded close together, so their own volume cannot be neglected and the gas is less compressible than an ideal gas.

Question: 4. What is meant by diffusion and effusion? Explain Graham's law of diffusion.

Answer: Diffusion is the spontaneous mixing of gases (or particles) due to their random molecular motion, so that they spread uniformly throughout the available space. Effusion is the escape of a gas through a tiny hole or porous material into a region of lower pressure. Graham's law states that the rate of diffusion (or effusion) of a gas is inversely proportional to the square root of its density or molar mass at constant temperature and pressure. Thus for two gases, r1/r2 = sqrt(M2/M1); lighter gases like hydrogen diffuse much faster than heavier gases like oxygen.

## Numerical Questions

Question: 1. Gorakh hill station is coldest area of Sindh province and has 83Kpa barometer pressure. What will be the pressure of this area in psi and atm units?

Answer: Given pressure = 83 kPa. In atmospheres: 1 atm = 101.325 kPa, so P = 83/101.325 = 0.819 atm. In psi: 1 atm = 14.7 psi, so P = 0.819 x 14.7 = about 12.04 psi. Hence the barometric pressure at Gorakh hill station is about 0.82 atm, or about 12.0 psi.

Question: 2. Calculate the volume occupied by 8g of methane gas at 40 degree C and 842 torr pressure? [Ans: V = 11.58 dm3]

Answer: Moles of CH4 = 8/16 = 0.5 mol. Pressure = 842/760 = 1.108 atm, and T = 40 + 273 = 313K. Using the general gas equation V = nRT/P = (0.5 x 0.0821 x 313)/1.108 = 11.58 dm3. So 8g of methane occupies 11.58 dm3 under the given conditions.

Question: 3. At 35 degree C, oxygen gas in a cylinder has 456 cm3 volume and 0.85 atm pressure. Calculate the pressure when this oxygen gas is transferred to 10 dm3 cylinder and cooled to 20 degree C. [Ans: 0.037 atm]

Answer: Convert V1 = 456 cm3 = 0.456 dm3, T1 = 308K, T2 = 293K, V2 = 10 dm3. Using the combined gas law P2 = P1V1T2/(V2T1) = (0.85 x 0.456 x 293)/(10 x 308) = 0.037 atm. The pressure falls greatly because the gas is expanded into a much larger volume.

Question: 4. Compare the rates of diffusion of the following pairs of gases: (1) H2 and D2 (2) He and SO2 (3) SF6 and SO2 [Ans: 1.41/1, 4/1, 1.51/1]

Answer: By Graham's law, rate ratio = sqrt(M of heavier gas / M of lighter gas). (1) H2 (M = 2) and D2 (M = 4): r(H2)/r(D2) = sqrt(4/2) = 1.41, i.e. 1.41/1. (2) He (M = 4) and SO2 (M = 64): r(He)/r(SO2) = sqrt(64/4) = 4, i.e. 4/1. (3) SF6 (M = 146) and SO2 (M = 64): r(SO2)/r(SF6) = sqrt(146/64) = 1.51, i.e. the lighter SO2 diffuses 1.51 times faster, ratio 1.51/1.

Question: 5. Four containers of equal volume are filled as follows: (i) 2.0g of H2 at 0 degree C (ii) 1.0g of H2 at 273 degree C (iii) 24g of O2 at 0 degree C (iv) 16g of CH4 at 273 degree C (a) Which container is at the greatest pressure? (b) Which container is at the lowest pressure? [Ans: a = iv and b = iii]

Answer: For equal volumes, pressure is proportional to n x T (from PV = nRT). (i) n = 1 mol, T = 273K, nT = 273. (ii) n = 0.5 mol, T = 546K, nT = 273. (iii) n = 24/32 = 0.75 mol, T = 273K, nT = 204.75. (iv) n = 1 mol, T = 546K, nT = 546. (a) Container (iv) has the greatest nT, so it is at the greatest pressure. (b) Container (iii) has the smallest nT, so it is at the lowest pressure.

Question: 6. A 500cm3 vessel contains H2 gas at 400 torr pressure and another 1dm3 vessel contains O2 gas at 600 torr pressure. If under the similar condition of temperature these gases are transferred to 2dm3 empty vessel, calculate the pressure of the mixture of gases in new vessel.

Answer: On transfer, each gas expands into 2 dm3, so find its partial pressure by Boyle's law. For H2: P = 400 x 0.5/2 = 100 torr. For O2: P = 600 x 1/2 = 300 torr. By Dalton's law, total pressure of the mixture = 100 + 300 = 400 torr. Hence the mixture in the 2 dm3 vessel has a pressure of 400 torr.

Question: 7. If 16 cm3 of hydrogen effuses in 30 sec, from a porous material, what volume of SO2 will effuse in the same time (30 sec.) under similar conditions?

Answer: By Graham's law, volume effused in the same time is inversely proportional to the square root of molar mass. V(SO2) = V(H2) x sqrt(M of H2 / M of SO2) = 16 x sqrt(2/64) = 16 x sqrt(1/32) = 16/5.66 = 2.83 cm3. So only about 2.83 cm3 of SO2 will effuse in 30 seconds, because SO2 is much heavier than hydrogen.

Question: 8. 40dm3 of hydrogen gas was collected over water at 831 torr pressure at 23 degree C. What would be the volume of dry hydrogen gas

at standard conditions? The vapour pressure of water at 23 degree C is 21 torr of Hg.

Answer: Pressure of dry H2 = 831 - 21 = 810 torr. T1 = 23 + 273 = 296K; at standard conditions P2 = 760 torr and T2 = 273K. Using the combined gas law, V2 = P1V1T2/(T1P2) = (810 x 40 x 273)/(296 x 760) = about 39.3 dm3. Hence the volume of dry hydrogen at standard conditions is about 39.3 dm3.

# Chapter 5: State of Matter II: Liquids

Multiple Choice Questions: 1. Choose the correct answer

(i) The boiling point of water (H2O) is 100 degree C where as that of hydrogen sulphide (H2S) is 42 degree C, This can be attributed to:

(a) Smaller bond angle of H2S than H2O

(b) Smaller radii of oxygen than sulphur

(c) High I.P of oxygen than sulphur

✓ (d) Tendency of water to form hydrogen bond

Reason: Water molecules are held together by strong hydrogen bonds, which need extra heat to break, so water has a much higher boiling point than H2S, which has no hydrogen bonding.

(ii) Liquids can form convex meniscus in a narrow glass tube when:

✓ (a) Cohesive forces are stronger than adhesive forces

(b) Adhesive forces are stronger than cohesive forces

(c) Cohesive and adhesive forces are equal in strength

(d) None of these

Reason: A convex meniscus (as with mercury) forms when the cohesive forces within the liquid are stronger than the adhesive forces between the liquid and the glass.

Question: (iii) Which of the following statement is incorrect? (a) Viscosity is the resistance against flow of liquid (b) Liquid possess definite volume (c) One feels sense of cooling after bath due to condensation (d) Sublimation is endothermic process.

Answer: (c) — The cooling felt after a bath is due to evaporation of water from the skin, not condensation, so statement (c) is incorrect.

(iv) Which of the following possesses weakest London dispersion forces:

(a) Cl2

✓ (b) F2

(c) Br2

(d) I2

Reason: F2 is the smallest halogen molecule with the fewest electrons, so its temporary dipoles are weakest and it has the weakest London dispersion forces.

Question: (v) Hydrogen bond is not found in: (a) H2O (b) CH4 (c) NH3 (d) HF

Answer: (d) — As given in the book's answer key. (Note: scientifically, hydrogen bonding is absent in CH4 (b), because carbon is not a highly electronegative atom; HF in fact shows the strongest hydrogen bonding.)

(vi) The boiling points of different liquids may be different at the same external pressure due to:

(a) Amounts of liquids

✓ (b) Intermolecular forces

(c) Surface area

(d) Viscosities

Reason: Boiling point depends on the strength of intermolecular forces in the liquid; stronger forces give a higher boiling point at the same external pressure.

Question: (vii) Which statement is incorrect about evaporation? (a) It is an exothermic process (b) It is a reverse process of condensation (c) It occurs at all temperature and pressure (d) It causes cooling effect

Answer: (a) — Evaporation absorbs heat from the surroundings, so it is an

endothermic process, not exothermic; statement (a) is therefore incorrect.

(viii) Which of the following liquids show maximum surface tension?

(a) Water

✓ (b) Mercury

(c) Ethyl alcohol

(d) Gasoline

Reason: Mercury has very strong cohesive (metallic) forces between its atoms, so it shows the maximum surface tension among the given liquids.

(ix) A non-polar molecule with bigger size will experience:

✓ (a) London forces

(b) dipole-dipole interaction

(c) Hydrogen bonding

(d) All of these

Reason: A non-polar molecule cannot have dipole-dipole forces or hydrogen bonding; it experiences only London dispersion forces, which increase with molecular size.

(x) Cooking time is reduced in a pressure cooker because:

✓ (a) Boiling point of water increases

(b) Boiling point of water decreases

(c) Vapor pressure of liquid is reduced

(d) Heat is uniformly distributed

Reason: In a pressure cooker the increased pressure raises the boiling point of water above 100 degree C, so food cooks faster at the higher temperature.

## Short Questions

Question: 1. Explain the following in terms of kinetic molecular interpretation of liquids: (a) Diffusion (b) Compression (c) Expansion (d) Molecular motion

Answer: (a) Diffusion: liquid molecules are in constant motion and slowly intermingle with the molecules of another liquid, though diffusion is slower than in gases because the molecules are closer together. (b) Compression: liquid molecules are already closely packed with very little empty space, so liquids are almost incompressible. (c) Expansion: on heating, the kinetic energy of the molecules increases and they move slightly apart, so liquids expand much less than gases. (d) Molecular motion: liquid molecules do not have fixed positions; they slide over one another, which is why liquids can flow and take the shape of the container.

Question: 2. Name three major kinds of intermolecular forces in liquids. Explain intermolecular forces in HCl.

Answer: The three major kinds of intermolecular forces in liquids are dipole-dipole interactions, London dispersion forces, and hydrogen bonding. HCl is a polar molecule with a permanent dipole, positive on hydrogen and negative on chlorine. In liquid HCl the positive end of one molecule attracts the negative end of a neighbouring molecule, giving dipole-dipole interactions. In addition, weak London dispersion forces act between all the molecules due to temporary dipoles. Hydrogen bonding of the strong type (as in water) is absent in HCl because chlorine is large and less electronegative in effect.

Question: 3. Define the following: (a) Molar heat of fusion (b) Molar heat of vapourization (c) Molar heat of Sublimation

Answer: (a) Molar heat of fusion is the amount of heat required to convert one mole of a solid completely into liquid at its melting point; for water it

is +6.02 KJ/mol. (b) Molar heat of vapourization is the heat required to convert one mole of a liquid into its vapours at its boiling point; for water it is 40.7 KJ/mol. (c) Molar heat of sublimation is the heat absorbed when one mole of a solid changes directly into vapours without passing through the liquid state. It is equal to the sum of the molar heat of fusion and the molar heat of vapourization.

Question: 4. Describe hydrogen bonding in water and explain the anomalous behavior of water due to hydrogen bonding.

Answer: In water, the hydrogen atom of one molecule, being partially positive, is attracted to a lone pair of the oxygen atom of another molecule, forming a hydrogen bond. Each water molecule can form up to four hydrogen bonds, building an open, cage-like structure in ice. Because of this open structure, ice is less dense than liquid water and floats on it. Water also expands below 4 degree C, so its density is maximum at 4 degree C; this anomalous behaviour protects aquatic life, as ice forms only at the surface of ponds in winter.

Question: 5. Water is more volatile than glycerin but petrol is more volatile than water at the same temperature. Explain in term of intermolecular force.

Answer: Volatility increases as intermolecular forces become weaker, because molecules escape more easily from the surface. Glycerin has three -OH groups per molecule and very extensive hydrogen bonding, so it is much less volatile than water, which has weaker and fewer hydrogen bonds per molecule. Petrol consists of non-polar hydrocarbon molecules held only by weak London forces, so its molecules escape even more easily than water molecules. Hence the order of volatility is petrol > water > glycerin, exactly following the decreasing strength of intermolecular

forces.

Question: 6. Give reasons for the following: i. Water spilled on floor evaporate faster than the same amount of water in a container. ii. A falling drop of a liquid is spherical. iii. Evaporation is a cooling process. iv. Boiling point of liquid remains constant although heat is continuously supplied to the liquid. v. Mercury has its meniscus upward. vi. Liquids cannot be compressed as gases do. vii. Density of water is highest at 4 degree C. viii. Honey is more viscous than water.

Answer: i. Spilled water has a much larger surface area, and evaporation takes place from the surface, so it evaporates faster. ii. Surface tension pulls the surface of a drop into the smallest possible area, which is a sphere. iii. The molecules with the highest kinetic energy escape during evaporation, lowering the average energy and temperature of the remaining liquid. iv. At the boiling point the supplied heat is used to overcome intermolecular forces (latent heat) instead of raising the temperature. v. In mercury cohesive forces are stronger than adhesive forces with glass, so the surface curves upward (convex meniscus). vi. Liquid molecules are already closely packed with little empty space, unlike gas molecules. vii. Below 4 degree C hydrogen bonding builds an open structure that expands water, while above 4 degree C normal thermal expansion dominates, so density is maximum at 4 degree C. viii. Honey contains large sugar molecules with extensive hydrogen bonding, giving strong internal friction (viscosity), while water molecules are small and flow easily.

## Descriptive Questions

Question: 1. What is Surface tension? Explain it with example. Give its unit and describe the factors that affect on surface tension of liquids?

Answer: Surface tension is the force responsible for the tension at the surface of a liquid; it is the energy required to stretch (increase) the surface by unit area. The surface molecules are pulled inwards by the molecules below them, so the surface behaves like a stretched elastic membrane; for example, a needle can float on water and insects can walk on it. Its unit is N/m (or J/m2). Surface tension decreases with rise in temperature and decreases when detergents or impurities are added, while strong intermolecular forces, such as hydrogen bonding and metallic bonding, give high surface tension as in water and mercury.

Question: 2. Explain the factors why honey is more viscous than water?

Answer: Viscosity is the internal resistance to the flow of a liquid, caused by friction between its layers. Honey is more viscous than water because its molecules (large sugar molecules such as glucose and fructose) are big, irregular and strongly attracted to each other by extensive hydrogen bonding. These large molecules entangle and cannot slide past one another easily, so honey flows very slowly. Water molecules are small, and although they also hydrogen bond, the layers slip over each other with much less resistance. Viscosity also decreases on heating, which is why warm honey flows faster than cold honey.

Question: 3. What is hydrogen bond? How is it established? Give its application in industrial and biochemical process.

Answer: A hydrogen bond is the force of attraction between a partially positively charged hydrogen atom in one molecule and a lone pair of a highly electronegative atom (F, O or N) in another molecule. It is established when hydrogen is covalently bonded to such an electronegative atom, which pulls the shared electrons towards itself and leaves the hydrogen with a partial positive charge that attracts a

neighbouring lone pair. In industry, hydrogen bonding explains the high solubility of substances like sugar and alcohol in water and is important in making adhesives, paints and fibres such as nylon. In biochemical processes it holds together the two strands of DNA, gives proteins their shapes, and is essential in enzyme action and the working of medicines.

Question: 4. What is meant by liquid crystals? How is it differing from liquids and crystalline solids?

Answer: Liquid crystals are a state of matter between liquids and crystalline solids, in which the molecules keep an ordered arrangement like a crystal but can still flow like a liquid. They differ from ordinary liquids because their molecules are arranged in a regular direction, which makes them show optical properties such as double refraction. They differ from crystalline solids because they have no rigid three-dimensional lattice, melt over a range instead of a sharp melting point, and can flow and change shape. Liquid crystals are widely used in LCD displays of watches, calculators, ATM machines and computer screens.

# Chapter 6: State of Matter III: Solids

## Multiple Choice Questions: 1. Choose the correct answer

(i) Quantity of energy released when 1 mole of the ionic crystal is formed from the gaseous ions is called:

(a) Bond energy

(b) Potential energy

(c) Ionization energy

✓ (d) Lattice energy

Reason: By definition, lattice energy is the energy released when one mole of an ionic crystal is formed from its isolated gaseous ions.

(ii) Graphite can conduct electricity in one direction only. With this behavior, it is said to be:

(a) Amorphous

(b) Symmetrical

✓ (c) Anisotropic

(d) Polymorphous

Reason: A solid whose physical properties (here, electrical conduction) differ in different directions is called anisotropic; graphite conducts along its layers but not across them.

(iii) NaCl and MgO are isomorphous solids because these have same:

(a) Melting point

(b) Boiling point

✓ (c) Geometrical shapes

(d) Chemical Properties

Reason: Isomorphous substances have the same crystalline form and

geometrical shape; NaCl and MgO both crystallize in the same cubic (rock-salt) structure.

(iv) The temperature at which two allotropic forms co-exist in equilibrium is called as:

(a) Melting temperature

(b) Fusion temperature

✓ (c) Transition temperature

(d) Critical temperature

Reason: The transition temperature is the temperature at which two allotropic forms of an element co-exist in equilibrium, e.g. rhombic and monoclinic sulphur at 96 degree C.

(v) Diamond is the example of:

(a) Ionic solid

✓ (b) Covalent solid

(c) Metallic solid

(d) Molecular solid

Reason: In diamond every carbon atom is joined to four others by covalent bonds in a giant network, so diamond is a covalent (network) solid.

(vi) Ice is a molecular solid, the intermolecular force of attraction among its molecules are:

(a) Ionic bond

(b) Covalent bond

(c) Metallic bond

✓ (d) Hydrogen bond

Reason: Ice consists of separate water molecules (a molecular solid),

and the molecules are held together by hydrogen bonds.

(vii) If all three axes in unit cell are of different length but all angles are of 90 degree then it is identified as:

✓ (a) Orthorhombic system

(b) Cubic system

(c) Tetragonal system

(d) Hexagonal system

Reason: A unit cell with a not equal to b not equal to c and all angles equal to 90 degree belongs to the orthorhombic system.

(viii) In NaCl, each Na+ ion is surrounded by Cl- ions in the numbers:

(a) Four

(b) Three

✓ (c) Six

(d) Seven

Reason: In the NaCl structure each Na+ ion is surrounded octahedrally by six Cl- ions, and each Cl- ion by six Na+ ions (6:6 coordination).

(ix) It is an example of crystalline solid:

✓ (a) Alum

(b) Glass

(c) Plastic

(d) Rubber

Reason: Alum has a definite geometrical shape, sharp melting point and an ordered lattice, so it is a crystalline solid; glass, plastic and rubber are amorphous.

(x) Each unit cell of sodium chloride consists of:

(a) Eight Na+ and Eight Cl- ions

(b) Two Na+ and Two Cl- ions

(c) Four Na+ and Four Cl- ions

(d) Four Na+ and Six Cl- ions

Reason: Counting corner, edge and face contributions, each NaCl unit cell contains four Na+ ions and four Cl- ions, i.e. four formula units.

## Short Questions

Question: 1. Define the following terms: * Allotropy * Sublimation * Melting point * Unit cell

Answer: Allotropy is the phenomenon in which the same element exists in two or more different crystal structures (forms) in the same physical state, e.g. diamond and graphite. Sublimation is the direct conversion of a solid into gas (vapours) on heating, without passing through the liquid state, shown by solids with weak intermolecular forces like iodine and naphthalene. Melting point is the fixed temperature at which a solid changes into liquid at atmospheric pressure. A unit cell is the smallest building block (repeating unit) of a crystal, by repetition of which the whole crystal lattice is built.

Question: 2. Define anisotropy. Which type of solids have this property?

Answer: Anisotropy is the property of a substance of showing different values of certain physical properties, such as electrical conduction, thermal expansion and refractive index, when measured in different directions. Crystalline solids show anisotropy because their particles are arranged in a regular order, and the arrangement is different along different directions. For example, graphite conducts electricity parallel to its layers but not perpendicular to them. Amorphous solids are isotropic because their random arrangement is the same in all directions.

Question: 3. What is Transition temperature? Explain with examples.

Answer: The transition temperature is the temperature at which two allotropic forms of the same element co-exist in equilibrium; below and above it, one or the other form is more stable. For example, rhombic sulphur is stable below 96 degree C and monoclinic sulphur above it, so 96 degree C (369K) is the transition temperature of sulphur, and both forms co-exist there. Another example is tin: white (metallic) tin changes into grey (powdery) tin below 13.2 degree C, which is its transition temperature. At the transition temperature the two forms have equal stability.

Question: 4. Explain why diamond is non-conductor of electricity and Graphite is a conductor?

Answer: In diamond, each carbon atom uses all four of its valence electrons in covalent bonds with four neighbouring atoms, so there are no free electrons to carry current; hence diamond is a non-conductor. In graphite, each carbon atom is bonded to only three others in flat layers, and the fourth electron of each atom is spare (delocalized) over the layer. These free electrons can move along the layers when a voltage is applied. Therefore graphite conducts electricity, but only parallel to its layers, making it anisotropic.

Question: 5. Define lattice energy. Explain how it is effected by size and the charge of ion.

Answer: Lattice energy is the energy released when one mole of an ionic crystal is formed from its isolated gaseous ions; it is a measure of the strength of the ionic lattice. It increases when the size of the ions decreases, because smaller ions come closer together and attract each other more strongly, e.g. LiF has a higher lattice energy than NaBr. It also

increases greatly with the charge on the ions, since the electrostatic attraction is proportional to the product of the charges; thus MgO (with +2 and -2 ions) has a much higher lattice energy than NaCl (with +1 and -1 ions).

Question: 6. Differentiate between the following: (i) Crystalline and amorphous solids (ii) Isomorphism and polymorphism

Answer: (i) Crystalline solids have particles arranged in a definite, regular lattice, a sharp melting point, definite geometrical shapes and anisotropic behaviour, e.g. NaCl and alum. Amorphous solids have randomly arranged particles, melt over a range of temperature, have no definite shape and are isotropic, e.g. glass and rubber. (ii) Isomorphism is the phenomenon in which different solid substances exist in the same crystalline structure, e.g. NaCl and MgO. Polymorphism is the phenomenon in which one solid substance exists in more than one crystalline shape under different conditions, e.g. sulphur as rhombic and monoclinic forms.

Question: 7. Explain the following with scientific reason. (i) Why the compressibility of solids is nearly zero? (ii) Why some solids are sublime in nature?

Answer: (i) In solids the particles are already tightly packed together with very small empty spaces and strong attractive forces between them. There is almost no room for the particles to be pushed closer, so the compressibility of solids is nearly zero. (ii) Some solids, like iodine, naphthalene and camphor, have very weak intermolecular forces between their particles. On heating, their surface particles gain enough energy to escape directly into the vapour state without melting first. Such solids are said to be sublime in nature.

## Descriptive Questions

Question: 1. Define unit cell. Draw a diagram to show its axial distances and axial angles.

Answer: A unit cell is the smallest portion (building block) of a crystal lattice which, when repeated over and over again in three dimensions, builds up the whole crystal. It is described by three axial distances (edge lengths) a, b and c, measured along the three axes, and three axial angles: alpha between the b and c axes, beta between the a and c axes, and gamma between the a and b axes. On the basis of these parameters, crystals are classified into seven crystal systems, such as cubic (a = b = c, all angles 90 degree) and orthorhombic. (As instructed, the diagram is described in words: a parallelepiped with edges a, b, c and opposite angles alpha, beta, gamma.)

Question: 2. Give four properties of ionic solids. How can you determine the number of Na+ and Cl- ions in a unit cell of sodium chloride?

Answer: Four properties of ionic solids are: they are hard and brittle because of strong electrostatic forces between oppositely charged ions; they have high melting and boiling points; they do not conduct electricity in the solid state but conduct when molten or in aqueous solution, because the ions become free to move; and they are soluble in polar solvents like water but insoluble in non-polar solvents. In the NaCl unit cell, the Cl- ions occupy the 8 corners (8 x 1/8 = 1) and the 6 face centres (6 x 1/2 = 3), giving 4 Cl- ions. The Na+ ions occupy the 12 edges (12 x 1/4 = 3) and the body centre (1), giving 4 Na+ ions. Thus each unit cell contains 4 Na+ and 4 Cl- ions.

Question: 3. Give a brief account on metallic solids. Why they conduct heat and electricity?

Answer: Metallic solids consist of positive metal ions (kernels) arranged in a regular lattice, surrounded by a sea of free, delocalized valence electrons; the attraction between the kernels and this electron sea is the metallic bond. They are generally hard, lustrous, malleable and ductile, and have high melting points and densities. They conduct electricity because the free electrons can move through the lattice and carry charge when a voltage is applied. They conduct heat because these mobile electrons quickly transfer kinetic energy from the hot part to the cold part of the metal. Examples are iron, copper, aluminium and sodium.

Question: 4. What is meant by polymorphism? How it is related to allotropy. Give your answer with examples.

Answer: Polymorphism is the phenomenon in which a solid substance exists in more than one crystalline shape (form) under different conditions of temperature and pressure. Allotropy is the same phenomenon applied to an element: when an element exists in two or more crystalline forms, the forms are called allotropes. Thus allotropy is a special case of polymorphism for elements. For example, carbon shows allotropy as diamond and graphite; sulphur is polymorphic (allotropic) as rhombic and monoclinic sulphur, changing at the transition temperature of 96 degree C. Compounds can also be polymorphic, e.g. calcium carbonate as calcite and aragonite.

Question: 5. Molecular solids are soft and possess low melting point, how can you justify this statement?

Answer: Molecular solids consist of individual molecules held together only by weak intermolecular forces such as London forces, dipole-dipole attractions or hydrogen bonds, while the covalent bonds inside each molecule are strong. Very little energy is needed to overcome these weak

forces, so the solid breaks up easily on heating and has a low melting point. The weak forces also cannot resist a deforming pressure, so molecular solids are soft and compressible. Examples justify this: ice, solid $CO_2$ (dry ice) and naphthalene are all soft solids with low melting or sublimation points, unlike ionic or covalent network solids.

# Chapter 7: Chemical Equilibrium

## Multiple Choice Questions: 1. Choose the correct answer

(i) In the equilibrium system of PCl5(g) <-> PCl3(g) + Cl2(g), the correct relationship between Kc and Kp is:

✓ (a) Kp > Kc

(b) Kp < Kc

(c) Kp = Kc

(d) Kp/Kc = 1

Reason: Here Dn (change in gas moles) = 2 - 1 = +1, and Kp = Kc(RT)^Dn = Kc(RT), so Kp is greater than Kc.

(ii) The equilibrium of which of the following reaction would not be affected by an increase in pressure:

Reason: In N2 + O2 <-> 2NO the number of gas moles is the same on both sides (2 = 2, Dn = 0), so a pressure change does not shift the equilibrium.

(iii) What happens to the value of Kc when a catalyst is added to a chemical system at equilibrium?

(a) It decreases

(b) It increases

(c) It becomes zero

✓ (d) It remains unchanged

Reason: A catalyst speeds up the forward and backward reactions equally and only helps equilibrium to be reached sooner; it does not change the value of Kc.

(iv) The correct Ksp expression of a sparingly soluble salt Li2C2O4 among the following is:

(a) Ksp = [Li+][C2O4 2-]

✓ (b) Ksp = [Li+]^2 [C2O4 2-]

(c) Ksp = [2Li+][C2O4 2-]

(d) Ksp = [2Li+]^2 [C2O4 2-]^2

Reason: Li2C2O4 dissociates into 2Li+ and one C2O4 2-, so Ksp = [Li+]^2 [C2O4 2-], each concentration raised to its stoichiometric coefficient.

(v) If the equilibrium expression of a reversible reaction is Kc = [C]^2 / [A][B] The balanced equilibrium equation should be:

(a) C <-> A + B

(b) A + B <-> C

(c) 2C <-> A + B

✓ (d) A + B <-> 2C

Reason: The products appear in the numerator with powers equal to their coefficients; [C]^2 over [A][B] corresponds to the equation A + B <-> 2C.

(vi) The equilibrium of N2(g) + O2(g) <-> 2NO(g) (DH = +ve) is affected by change in:

✓ (a) Temperature only

(b) Pressure only

(c) Both temperature and pressure

(d) Neither temperature nor pressure

Reason: Dn = 0, so pressure has no effect; but the reaction is endothermic, so a temperature change shifts the equilibrium and changes Kc.

(vii) In the reaction A(g) + B(g) <-> 2C(g), the equilibrium constants Kp = Kc because:

(a) Dn > 1

(b) Dn < 1

(c) Dn = 1

✓ (d) Dn = 0

Reason: Kp = Kc(RT)^Dn; when Dn = 0 (2 moles of gas on each side), (RT)^0 = 1 and Kp = Kc.

(viii) The term active mass use in law of mass action means:

(a) No. of mole

(b) No. of molecules

✓ (c) mole per dm3

(d) gram per dm3

Reason: Active mass means the molar concentration of a substance, i.e. number of moles per dm3 of solution (or partial pressure for a gas).

(ix) NaCl when added to an aqueous silver chloride solution:

✓ (a) Decreases the solubility of AgCl

(b) Increases the solubility of AgCl

(c) Forms a clear solution

(d) Does not effect

Reason: NaCl supplies the common ion Cl-, which shifts the equilibrium AgCl(s) <-> Ag+ + Cl- to the left (common ion effect), decreasing the solubility of AgCl.

(x) Some reactions are nearly to completion in the forward direction and identified by their:

(a) Very high value of Kc

(b) Very low value of Kc

(c) Very high value of DH

(d) Very low value of DH

Reason: A very large Kc means the equilibrium mixture contains mostly products, so the reaction goes nearly to completion in the forward direction.

## Short Questions

Question: 1. Define the following: (i) Reversible reaction (ii) Chemical equilibrium

Answer: (i) A reversible reaction is a reaction which can proceed in both the forward and the backward direction under the same conditions, and is written with a double arrow; for example, N2 + 3H2 <-> 2NH3. Such reactions never go to completion because the products react to reform the reactants. (ii) Chemical equilibrium is the state of a reversible reaction at which the rate of the forward reaction becomes equal to the rate of the backward reaction. At equilibrium there is no net change in the concentrations of reactants and products with time, although both reactions continue at the microscopic level, so it is a dynamic equilibrium.

Question: 2. Write expression of Kc and Kp for the following reversible reactions. (i) 2NO2(g) + 7H2(g) <-> 2NH3(g) + 4H2O(g) (ii) 2H2S(g) + 3O2(g) <-> 2H2O(g) + 2SO2(g)

Answer: (i) Kc = [NH3]^2 [H2O]^4 / ([NO2]^2 [H2]^7), and Kp = (P NH3)^2 (P H2O)^4 / ((P NO2)^2 (P H2)^7). (ii) Kc = [H2O]^2 [SO2]^2 / ([H2S]^2 [O2]^3), and Kp = (P H2O)^2 (P SO2)^2 / ((P H2S)^2 (P O2)^3). In each expression the product concentrations are written in the numerator and the reactant concentrations in the denominator, each

raised to the power of its coefficient in the balanced equation. Kp uses partial pressures in place of concentrations for gaseous reactions.

Question: 3. (a) Define solubility product. (b) Write the solubility product expression of the following salts. BaF2, Li2C2O4, MgCO3, Ag3PO4

Answer: (a) The solubility product (Ksp) is the product of the molar concentrations of the ions of a sparingly soluble salt in its saturated solution at a given temperature, each concentration raised to the power of its coefficient in the dissociation equation. (b) BaF2 <-> Ba2+ + 2F-: Ksp = [Ba2+][F-]^2. Li2C2O4 <-> 2Li+ + C2O4 2-: Ksp = [Li+]^2 [C2O4 2-]. MgCO3 <-> Mg2+ + CO3 2-: Ksp = [Mg2+][CO3 2-]. Ag3PO4 <-> 3Ag+ + PO4 3-: Ksp = [Ag+]^3 [PO4 3-].

Question: 4. Using Le-Chatelier's principle, explain three ways in which yield of SO3 can be increase in Contact process. 2SO2(g) + O2(g) <-> 2SO3(g) (DH = -ve)

Answer: The reaction is exothermic and shows a decrease in gas moles (3 moles give 2 moles). First, increase the pressure: the equilibrium shifts towards the side with fewer moles, producing more SO3. Second, decrease the temperature: for an exothermic reaction, lowering the temperature favours the forward (heat-producing) direction, increasing SO3 (in practice a moderate 450 degree C is used with a V2O5 catalyst so the rate stays reasonable). Third, increase the concentration of SO2 or O2 (oxygen is used in excess), or remove SO3 as it is formed by absorbing it in concentrated H2SO4; both shift the equilibrium forward by Le-Chatelier's principle.

## Descriptive Questions

Question: 1. State Law of mass action and derive Kc expression of a

general reversible reaction.

Answer: The law of mass action states that, at constant temperature, the rate of a chemical reaction is proportional to the active masses (molar concentrations) of the reacting species. For the general reaction aA + bB <-> cC + dD, the forward rate = kf [A]^a [B]^b and the backward rate = kb [C]^c [D]^d. At equilibrium the two rates are equal, so kf [A]^a [B]^b = kb [C]^c [D]^d. Rearranging gives kf/kb = [C]^c [D]^d / ([A]^a [B]^b) = Kc, the equilibrium constant. Its value depends only on temperature and is fixed for a given balanced equation.

Question: 2. State Le-Chatelier's principle. Explain the industrial application of Le-Chatelier's principle using Haber's process.

Answer: Le-Chatelier's principle states that if a stress (change in concentration, pressure or temperature) is imposed on a system at equilibrium, the system shifts in the direction that reduces the effect of the applied stress. In Haber's process, N2 + 3H2 <-> 2NH3, DH = -92 KJ/mol; there are 4 moles of gas on the left and 2 on the right, and the reaction is exothermic. Therefore a high pressure (about 200 atm) shifts the equilibrium towards ammonia, and a lower temperature favours ammonia, but would make the rate too slow. In industry a compromise temperature of about 450 degree C and an iron catalyst are used, and ammonia is condensed and removed so the equilibrium keeps shifting forward.

Question: 3. Give brief account on Common ion effect.

Answer: The common ion effect is the decrease in the solubility (or ionization) of a sparingly soluble salt or weak electrolyte when a highly soluble electrolyte containing an ion common to it is added to its solution. For example, the solubility of AgCl decreases when NaCl is added, because

the extra Cl- ions shift the equilibrium AgCl(s) <-> Ag+ + Cl- to the left and more AgCl precipitates. Similarly, NH4Cl suppresses the ionization of weak NH4OH by supplying NH4+ ions. The effect is an application of Le-Chatelier's principle and is used in qualitative analysis, in salting out of soaps and in purification of salts.

Question: 4. Define equilibrium constant (Kc). How it helps in predicting the (i) Direction of reaction (ii) Extent of reaction.

Answer: The equilibrium constant Kc is the ratio of the product of molar concentrations of products to that of reactants at equilibrium, each raised to its stoichiometric coefficient; it has a fixed value at a given temperature. (i) Direction: the reaction quotient Qc is calculated with the actual concentrations; if Qc < Kc the reaction proceeds forward, if Qc > Kc it proceeds backward, and if Qc = Kc the system is at equilibrium. (ii) Extent: a large Kc (greater than 10^3) means products dominate and the reaction goes nearly to completion, a small Kc (less than 10^-3) means reactants dominate and the reaction hardly proceeds, while an intermediate value means appreciable amounts of both are present.

## Numerical Questions

Question: 1. At 444 degree C reaction of hydrogen and iodine is performed in a sealed 1dm3 steel vessel. H2(g) + I2(g) <-> 2HI(g) If equilibrium mixture contains 1 mole of H2, 1 mole of I2 and 7 moles of HI. Calculate (a) Equilibrium constant (Kc) (b) Initial concentration of H2 and I2 [Ans: Kc = 49, initial conc. of H2 = I2 = 4.5 moles]

Answer: The vessel is 1 dm3, so moles equal molar concentrations. (a) Kc = [HI]^2 / ([H2][I2]) = (7)^2 / (1 x 1) = 49. (b) 7 moles of HI were formed from 3.5 moles of H2 and 3.5 moles of I2 (2 moles HI come from 1 mole each of H2 and I2). Initial H2 = 1 + 3.5 = 4.5 moles, and initial I2 = 1 +

3.5 = 4.5 moles, i.e. initial concentrations of H2 and I2 are 4.5 mol/dm3 each.

Question: 2. Lead fluoride (PbF2) is a high melting white solid used in glass coating to reflect IR rays. Its solubility in water at 25 degree C is 0.58g/dm3. Calculate its Ksp. (At. mass of Pb = 207 and F = 19) [Ans: 5.3 x 10^-8 mol3/dm9]

Answer: Molar mass of PbF2 = 207 + 2(19) = 245 g/mol. Molar solubility s = 0.58/245 = 2.37 x 10^-3 mol/dm3. PbF2 <-> Pb2+ + 2F-, so [Pb2+] = s and [F-] = 2s. Ksp = [Pb2+][F-]^2 = s(2s)^2 = 4s^3 = 4 x (2.37 x 10^-3)^3 = 5.3 x 10^-8 mol3/dm9.

Question: 3. A solution of CaCO3 is prepared by mixing 200cm3 of 2.4 x 10^-4M Ca(NO3)2 and 300cm3 of 4.5 x 10^-2M K2CO3. Will CaCO3 precipitate upon cooling to 25 degree C if Ksp of CaCO3 at 25 degree C is 3.8 x 10^-9. [Ans: CaCO3 precipitate]

Answer: Total volume = 500 cm3. After mixing, [Ca2+] = 2.4 x 10^-4 x (200/500) = 9.6 x 10^-5 M, and [CO3 2-] = 4.5 x 10^-2 x (300/500) = 2.7 x 10^-2 M. Ionic product Qsp = [Ca2+][CO3 2-] = 9.6 x 10^-5 x 2.7 x 10^-2 = 2.6 x 10^-6. Since Qsp (2.6 x 10^-6) is greater than Ksp (3.8 x 10^-9), the solution is supersaturated and CaCO3 will precipitate.

Question: 4. Hydrogen iodide is a colourless gas prepared by reacting H2 and I2. H2(g) + I2(g) <-> 2HI(g) In an equilibrium study of above reaction, 1.2 mole of H2 and 1.2 moles of I2 were injected in an evacuated 1000cm3 sealed flask and start the reaction to occur at 440 degree C until the equilibrium is formed. If Kc for this reaction is 49, calculate the equilibrium concentration of H2, I2 and HI. [Ans: [H2] = 0.267 mol/dm3, [I2] = 0.267 mol/dm3, [HI] = 1.866 mol/dm3]

Answer: Flask volume = 1000 cm3 = 1 dm3, so initial [H2] = [I2] = 1.2 mol/dm3 and [HI] = 0. Let x mol/dm3 of H2 react; at equilibrium [H2] = [I2] = 1.2 - x and [HI] = 2x. Kc = (2x)^2/(1.2 - x)^2 = 49, so taking the square root, 2x/(1.2 - x) = 7, giving 2x = 8.4 - 7x and x = 0.933. Hence [H2] = [I2] = 1.2 - 0.933 = 0.267 mol/dm3, and [HI] = 2 x 0.933 = 1.866 mol/dm3.

# Chapter 8: Acids, Bases and Salts

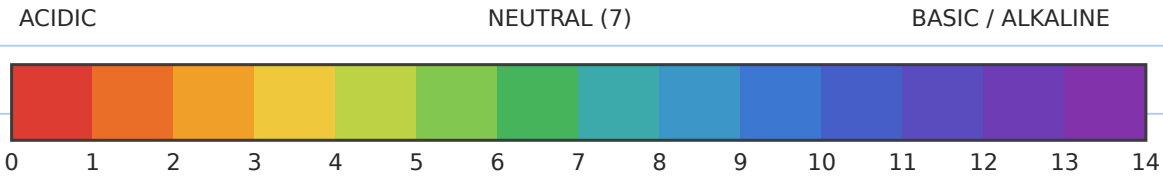



Fig: The pH Scale (0-14)

## Multiple Choice Questions: 1. Choose the correct answer

(i) $H_2SO_4$ is stronger acid than $CH_3COOH$ because:

(a) It gives two H+ ion per molecule

(b) Its boiling point is high

✓ (c) Its degree of ionization is high

(d) It is highly corrosive

Reason: Acid strength depends on the extent of ionization; $H_2SO_4$ ionizes almost completely in water while $CH_3COOH$ ionizes only slightly, so its degree of ionization is high.

(ii) $Al_2O_3$ is amphoteric oxide because it reacts with:

(a) Acids

(b) Base

✓ (c) Both acids and base

(d) neither acid nor base

Reason: An amphoteric oxide reacts with both acids and bases to form salt and water, as $Al_2O_3$ does with HCl and with NaOH.

(iii) Which of the following is not a Buffer solution:

(a) $CH_3COOH/CH_3COONa$

(b) $Na_2CO_3/NaHCO_3$

(c) NH4OH/NH4Cl

✓ (d) NaOH/HCl

Reason: NaOH/HCl is a mixture of a strong base and a strong acid, not a weak acid/base with its salt, so it cannot act as a buffer.

(iv) Which oxide is amphoteric in nature:

(a) K2O

(b) CO2

(c) CaO

✓ (d) Al2O3

Reason: Al2O3 reacts with both acids and bases, so it is amphoteric; K2O and CaO are basic oxides and CO2 is an acidic oxide.

(v) Which of the following does not alter the pH of a solution?

(a) NH4Cl

(b) Na2CO3

✓ (c) NaCl

Reason: NaCl is a salt of a strong acid and a strong base and does not hydrolyze, so its solution is neutral and it does not alter the pH.

(vi) Conjugated acid of NH3 is:

✓ (a) NH4+

(b) NH2-

(c) NH2

(d) NH

Reason: A conjugate acid is formed by gaining a proton; NH3 + H+ gives NH4+, so NH4+ is the conjugate acid of NH3.

(vii) Salt which is formed by the neutralization of weak acid and strong base is:

(a) $NaNO_3$

(b) $NH_4Cl$

✓ (c) $Na_2CO_3$

(d) $NH_4CN$

Reason: $Na_2CO_3$ is formed from weak carbonic acid ($H_2CO_3$) and strong base NaOH, so its solution is alkaline due to hydrolysis.

Question: (viii) Which of the following statements is not correct about the bases? (a) They have bitter tastes (b) They have high pH value (c) They react with acids to form salts (d) They turn blue litmus red

Answer: (d) — Bases turn red litmus blue, not blue litmus red (that is the action of acids), so statement (d) is not correct.

(ix) A conjugate acid base pair has the difference of only:

(a) One electron

✓ (b) One proton

(c) One electron pair

(d) One proton pair

Reason: Members of a conjugate acid-base pair differ from each other by only one proton ($H^+$), e.g. $NH_4^+$ and $NH_3$.

(x) Salt formed by neutralization of weak acid and weak base is:

(a) $NH_4Cl$

(b) $Na_2CO_3$

✓ (c) $NH_4CN$

(d) $Na_2SO_4$

Reason: $NH_4CN$ is formed from the weak acid HCN and the weak base $NH_4OH$, so it is the salt of a weak acid and a weak base.

## Short Questions

Question: 1. Define pH & pOH of a solution? Also show that pH + pOH = 14.

Answer: pH is the negative logarithm of the hydrogen ion concentration, pH = -log[H+]; it measures the acidity of a solution on a scale from 0 to 14. pOH is the negative logarithm of the hydroxide ion concentration, pOH = -log[OH-]. For water at 25 degree C, Kw = [H+][OH-] = 1 x 10^-14. Taking the negative log of both sides: -log[H+] - log[OH-] = 14, i.e. pH + pOH = 14. Hence a solution with pH less than 7 is acidic, equal to 7 neutral, and greater than 7 basic.

Question: 2. Why the aqueous solution of NH4Cl is acidic and Na2CO3 is alkaline.

Answer: NH4Cl is the salt of a weak base (NH4OH) and a strong acid (HCl). Its NH4+ ions hydrolyze with water to give H+ ions (NH4+ + H2O -> NH4OH + H+), so the solution becomes acidic. Na2CO3 is the salt of a strong base (NaOH) and a weak acid (H2CO3). Its CO3 2- ions hydrolyze to give OH- ions (CO3 2- + H2O -> HCO3- + OH-), so the solution becomes alkaline. In general, cation hydrolysis gives an acidic solution and anion hydrolysis gives a basic solution.

Question: 3. Write down conjugate base of each of the following acids. H2SO4, H2S, NH4+, HCOOH

Answer: A conjugate base is formed when an acid loses one proton. The conjugate base of H2SO4 is HSO4- (bisulphate ion). The conjugate base of H2S is HS- (bisulphide ion). The conjugate base of NH4+ is NH3 (ammonia). The conjugate base of HCOOH is HCOO- (formate ion). In each pair the acid and its conjugate base differ by exactly one proton, and a strong acid has a weak conjugate base.

Question: 4. What is meant by self-ionization of water? Write the expression of Kw. What is its value at 25 degree C?

Answer: Self-ionization of water is the process in which one water molecule donates a proton to another water molecule, producing hydronium and hydroxide ions: 2H2O <-> H3O+ + OH-. The ionic product of water is expressed as Kw = [H+][OH-] (more exactly [H3O+][OH-]). Its value at 25 degree C is 1 x 10^-14 mol2/dm6. In pure water [H+] = [OH-] = 1 x 10^-7 M, which is why pure water is neutral with pH 7 at this temperature.

Question: 5. Write equation and indicate the conjugate acid-base pairs for the following: (i) Acetic acid & ammonia (ii) Hydrochloric acid & water

Answer: (i) CH3COOH + NH3 <-> CH3COO- + NH4+. Here CH3COOH (acid) and CH3COO- (its conjugate base) form one pair, and NH4+ (conjugate acid) and NH3 (base) form the other pair; the proton passes from acetic acid to ammonia. (ii) HCl + H2O <-> Cl- + H3O+. Here HCl (acid) and Cl- (its conjugate base) form one pair, and H3O+ (conjugate acid) and H2O (base) form the other pair. In both reactions the acid and its conjugate base differ by one proton only.

## Descriptive Questions

Question: 1. Explain Bronsted-Lowry theory of acids and bases. What is meant by conjugate acid-base pair give examples?

Answer: According to the Bronsted-Lowry theory, an acid is a species which tends to donate a proton (H+), and a base is a species which tends to accept a proton. When an acid donates a proton, the species left behind can accept a proton back and is called its conjugate base; similarly, a base accepting a proton becomes its conjugate acid. A conjugate acid-base pair is two species that differ by only one proton, one on each side of the

equation. For example, in HCl + H2O <-> H3O+ + Cl-, HCl/Cl- and H3O+/H2O are conjugate pairs. Water can act as an acid with NH3 and as a base with HCl, so it is amphoteric in this theory.

Question: 2. Define the process of hydrolysis. Explain the behavior of each of the following salts in aqueous solution. (i) K2CO3 (ii) (NH4)2SO4 (iii) NaNO3

Answer: Hydrolysis is the reaction of the cation or anion of a salt with water, which changes the pH of the solution. (i) K2CO3 is a salt of a strong base and a weak acid; the CO3 2- ion hydrolyzes to produce OH- ions, so the solution is alkaline (pH > 7). (ii) (NH4)2SO4 is a salt of a weak base and a strong acid; the NH4+ ion hydrolyzes to produce H+ ions, so the solution is acidic (pH < 7). (iii) NaNO3 is a salt of a strong base and a strong acid; neither Na+ nor NO3- hydrolyzes appreciably, so the solution remains neutral (pH = 7).

Question: 3. What is Buffer solution? Explain how it resists the change of pH by adding small amount of acid and base. Give the applications of buffer solution.

Answer: A buffer solution is a mixture of a weak acid with its salt of a strong base (acidic buffer), or a weak base with its salt of a strong acid (basic buffer), which resists a change in pH when a small amount of acid or base is added. In an acidic buffer like CH3COOH/CH3COONa, added H+ ions are removed by the acetate ions to form weakly ionized acetic acid, and added OH- ions are removed by the acetic acid to form acetate and water, so the pH stays nearly constant. Buffers are applied in maintaining the pH of blood (about 7.4), in fermentation and dyeing industries, in chemical analysis, and in medicines and shampoos. The range over which a buffer is effective is called its buffer range.

Question: 4. Describe Lewis theory of acids and bases. What are the advantages of this theory over Lowry Bronsted theory?

Answer: According to the Lewis theory, an acid is a substance that can accept an electron pair, and a base is a substance that can donate an electron pair. Thus BF3 is a Lewis acid because boron has an incomplete octet and accepts a lone pair from NH3, which is a Lewis base; the product is a coordinate (adduct) compound. Its advantages over the Bronsted-Lowry theory are: it is not limited to proton transfer or to aqueous solutions, so reactions like BF3 + NH3 and metal ion complex formation are included as acid-base reactions. It explains the acidic behaviour of oxides such as CO2 and SO3 and of metal ions. Hence it is the most general theory of acids and bases.

Question: 5. What is salt? Explain Acidic, Basic and Neutral salts?

Answer: A salt is an ionic compound formed by the neutralization reaction of an acid and a base, in which the H+ of the acid is replaced by a metal or ammonium ion, e.g. HCl + NaOH -> NaCl + H2O. Acidic salts still contain replaceable hydrogen and are formed by partial neutralization of a polybasic acid, e.g. NaHSO4 and NaHCO3. Basic salts still contain an OH group and are formed by partial neutralization of a polyacidic base, e.g. Mg(OH)Cl. Neutral (normal) salts are formed by complete neutralization and contain neither replaceable H nor OH, e.g. NaCl, Na2SO4 and CaCO3. Salts are also described by their solutions: salts of strong acid and weak base are acidic in solution, and vice versa.

## Numerical Questions

Question: 1. A solution is made by dissolving 14.8g HCl in water at 25 degree C. If the volume of solution is 750cm3 and HCl is assumed to be completely ionized, calculate its pH. [Ans: pH = 1.2]

Answer: Moles of HCl = 14.8/36.5 = 0.405 mol in 0.750 dm3, so [H+] = 0.405/0.750 = 0.54 mol/dm3, since HCl is completely ionized. pH = -log[H+] = -log(0.54) = 0.27. (Note: working with the given data gives pH about 0.27; the book prints the answer as pH = 1.2, which corresponds to about one-tenth of this concentration.)

Question: 2. The hydroxide ion concentration in an antiseptic solution at 25 degree C is 3.5 x 10^-4M. Calculate its pH. [Ans: pH = 8]

Answer: pOH = -log[OH-] = -log(3.5 x 10^-4) = 4 - log 3.5 = 4 - 0.54 = 3.46. Using pH + pOH = 14 at 25 degree C, pH = 14 - 3.46 = 10.54, so the solution is basic. (Note: the book prints the answer as pH = 8; the value calculated from the given hydroxide ion concentration is about 10.5.)

# Chapter 9: Chemical Kinetics

## Multiple Choice Questions: 1. Choose the correct answer

(i) For a hypothetical reaction x + y -> z, if the conc. of x is double, the rate increases by square and if the conc. of y is doubled the rate increases by twice. The experimental rate law of this reaction is:

(a) R = K [x]1 [y]1

(b) R = K [x]1 [y]2

✓ (c) R = K [x]2 [y]1

(d) R = K [x]2 [y]2

Reason: Doubling x makes the rate four times (square), so the order in x is 2; doubling y doubles the rate, so the order in y is 1, giving R = K [x]^2 [y]^1.

(ii) The unit of rate constant (K) for the first order reaction is:

✓ (a) s^-1

(b) conc. s^-1

(c) conc^-1. s

(d) conc.-1 s^-1

Reason: For a first order reaction, K = rate/concentration, so its units reduce to time^-1, i.e. s^-1.

(iii) Rate constant of a reaction is affected by:

(a) Conc. of reactants

(b) Conc. of products

✓ (c) Temperature

(d) Reaction time

Reason: The rate constant K depends only on temperature (and a

catalyst); it does not change with concentration or time.

(iv) The purpose of using of catalyst in a chemical reaction is to change:

(a) Equilibrium constant

(b) Enthalpy of reaction

✓ (c) Activation energy

(d) Nature of reaction

Reason: A catalyst provides an alternative path of lower activation energy, so the reaction becomes faster without changing the equilibrium constant or enthalpy.

(v) The overall order of reaction to which the rate law is R = K:

✓ (a) Zero order

(b) First order

(c) Second order

(d) Third order

Reason: R = K means the rate does not depend on any concentration (all powers are zero), so the reaction is of zero order.

(vi) Ionic reactions are classified into:

(a) Slow reaction

(b) Moderate reactions

✓ (c) Fast reactions

(d) Reversible reaction

Reason: Ionic reactions involve only the combination of ions already present in solution and need almost no bond breaking, so they are classified as fast (instantaneous) reactions.

(vii) The decomposition of $H_2O_2$ is inhibited by:

(a) 2% ethanol

(b) Glycerin

(c) $MnO_2$

(d) $V_2O_5$

Reason: Glycerin acts as a negative catalyst (inhibitor) and slows the decomposition of $H_2O_2$, whereas $MnO_2$ speeds it up.

(viii) The best alternative term for the velocity of reaction is:

(a) Rate of appearance

(b) Rate of disappearance

(c) Average rate

✓ (d) Instantaneous rate

Reason: Velocity of reaction at a particular moment is best expressed as the instantaneous rate, the rate measured over an extremely short time interval.

(ix) Order of reaction is the power to which concentration of reactant is:

(a) Lowered

✓ (b) Raised

(c) Stopped

(d) Constant

Reason: The order with respect to a reactant is the power to which its concentration is raised in the experimental rate law expression.

(x) In the reaction A -> B, the rate of disappearance is written as:

(a) dA/dt

✓ (b) -dA/dt

(c) dB/dt

(d) -dB/dt

Reason: The concentration of reactant A decreases with time, so the

✓

rate of disappearance is written with a negative sign, -dA/dt, to make the rate a positive quantity.

## Short Questions

Question: 1. Define the following: (a) Rate of reaction (b) Velocity of reaction (c) Order of reaction (d) Rate constant

Answer: (a) Rate of reaction is the decrease in concentration of a reactant or the increase in concentration of a product per unit time during a chemical reaction. (b) Velocity of reaction is the rate at a particular instant (instantaneous rate), measured from the slope of a concentration-time graph at that point. (c) Order of reaction is the sum of the powers to which the concentrations of the reactants are raised in the experimental rate law; it may be a whole number, a fraction or even zero. (d) Rate constant (specific rate constant) is the rate of reaction when the molar concentration of each reactant is unity; its value depends on temperature.

Question: 2. The rate law of the reaction 2NO + Br2 -> 2NOBr is given as Rate = K[NO]2 [Br2]. What is the order of reaction? (i) With respect to each reactant (ii) Overall reaction

Answer: (i) The power of [NO] in the rate law is 2, so the reaction is second order with respect to NO. The power of [Br2] is 1, so it is first order with respect to Br2. (ii) The overall order is the sum of these powers: 2 + 1 = 3, so the reaction is of third order overall. Note that the order is found experimentally from the rate law, not simply from the balancing coefficients, although here they agree.

Question: 3. How can you define order of reaction? What are the units for the rate constants of zero order, first order and second order reactions?

Answer: Order of reaction is the sum of the powers of the concentration

terms in the experimentally determined rate law; for Rate = K[A]^x [B]^y, the order is x + y. The units of K depend on the order. For a zero order reaction, K has the same units as rate: mol dm^-3 s^-1. For a first order reaction, K has units of s^-1. For a second order reaction, K has units of dm^3 mol^-1 s^-1.

Question: 4. Explains the following: (i) Powdered marble (CaCO3) reacts quickly with hydrochloric acid than solid lump of marble. (ii) Milk sour more rapidly in summer than in winter. (iii) Reactants in solution reacts faster at high concentration. (iv) Raising the temperature causes an increase in the rate of reaction.

Answer: (i) Powder has a much larger surface area than a lump, so more particles are exposed to the acid and collisions are more frequent. (ii) In summer the temperature is higher, so the bacteria that sour milk work faster and the reactions proceed at a greater rate. (iii) At high concentration there are more particles per unit volume, so effective collisions are more frequent and the rate increases. (iv) Raising the temperature increases the kinetic energy of molecules, so more of them can cross the activation energy barrier and the number of effective collisions rises sharply.

Question: 5. Differentiate between the following: (i) Rate of reaction and rate constant (ii) Homogenous and Heterogeneous catalyst (iii) Positive catalyst and inhibitor (iv) Elementary and overall reaction

Answer: (i) Rate of reaction is the change in concentration per unit time and changes as the reaction proceeds; the rate constant is the rate when each concentration is unity and stays constant at a fixed temperature. (ii) A homogeneous catalyst is in the same phase as the reactants, e.g. an acid catalyst in solution; a heterogeneous catalyst is in a different phase,

e.g. solid iron in Haber's process with gaseous reactants. (iii) A positive catalyst increases the rate by lowering the activation energy, while an inhibitor (negative catalyst) decreases the rate. (iv) An elementary reaction takes place in a single step, while an overall reaction is the sum of several elementary steps (its mechanism).

## Descriptive Questions

Question: 1. Enlist various factors which influence on the rate of chemical reactions and describe the effect of concentration and surface area of reactants on the reaction rate.

Answer: The rate of a chemical reaction is influenced by the nature of the reactants, concentration of reactants, surface area of solid reactants, temperature, light (radiations), molecular size and the presence of a catalyst. Effect of concentration: increasing the concentration puts more particles in the same volume, so collisions become more frequent and the rate increases; for many reactions the rate is proportional to a power of the concentration, as shown by the rate law. Effect of surface area: for a solid reactant, breaking it into small pieces or powder exposes more particles at the surface where reaction occurs. Hence powdered marble reacts with acid much faster than a lump of the same mass, and coal dust burns faster than coal lumps.

Question: 2. Explain in terms of collision theory how the reaction rate increases with the rise of temperature.

Answer: According to collision theory, a reaction occurs only when molecules collide with sufficient energy (at least the activation energy) and with proper orientation; such collisions are called effective collisions. When the temperature rises, the average kinetic energy of the molecules increases and they move faster, so the total number of collisions per

second increases. More importantly, the fraction of molecules having energy equal to or greater than the activation energy increases greatly. As a result, the number of effective collisions rises sharply and the reaction rate increases. This is why a rise of about 10 degree C often roughly doubles the rate of a reaction.

Question: 3. What is meant by energy of activation and activated complex? Explain with the help of potential energy diagram.

Answer: Activation energy is the additional (minimum) energy which the reacting molecules must possess, above their average kinetic energy, for their collision to be effective and produce products. As reactant molecules approach, their energy rises to a maximum, forming a high-energy, unstable and short-lived state called the activated complex (transition state), which then breaks down into products with release of energy. On a potential energy diagram, energy is plotted against the progress of the reaction: the curve rises from the reactants' level to a peak (the activated complex) and then falls to the products' level. The height of the peak above the reactants is the activation energy, and the difference between the product and reactant levels is the enthalpy change of the reaction.

Numerical Questions

Question: 1. Decomposition of NO2 into NO and O2 is of second order reaction. 2NO2 -> 2NO + O2 If the rate constant at certain temperature is 3.8 x 10^-4 dm3 mol^-1 s^-1 and the initial concentration of NO2 is 0.38M, calculate the initial rate of the reaction. [Ans: 5.48 x 10^-5 mol/dm3.s]

Answer: For a second order reaction, Rate = K [NO2]^2. Substituting: Rate = 3.8 x 10^-4 x (0.38)^2 = 3.8 x 10^-4 x 0.1444 = 5.48 x 10^-5 mol/dm3.s. So the initial rate of decomposition is 5.48 x 10^-5 mol dm^-3

s^-1.

Question: 2. The overall rate law for the reaction A + B -> C is R = K [A] [B] If in an experiment the initial concentration of A and B was found to be 0.43M and 0.78M respectively while the initial rate was 3.8 x 10^-3Ms^-1. Determine rate constant and mention its unit. [Ans: 1.13 x 10^-2 dm3/mol.s]

Answer: K = Rate / ([A][B]) = 3.8 x 10^-3 / (0.43 x 0.78) = 3.8 x 10^-3 / 0.3354 = 1.13 x 10^-2. The reaction is second order overall (first order in each reactant), so the unit of K is dm3/mol.s (dm3 mol^-1 s^-1). Hence the rate constant is 1.13 x 10^-2 dm3 mol^-1 s^-1.

Question: 3. The reaction 2NO + Cl2 -> 2NOCl was studied at 25 degree C. the following results were obtained [Experiment 1: NO = 0.1, Cl2 = 0.1, Initial rate = 2.52 x 10^-3; Experiment 2: NO = 0.1, Cl2 = 0.2, Initial rate = 5.04 x 10^-3; Experiment 3: NO = 0.2, Cl2 = 0.1, Initial rate = 10.05 x 10^-3 (concentrations in mol/dm3, rate in mol/dm3.s)] Illustrate the rate law and find the order of reaction. [Ans: Order of reaction = 3]

Answer: Compare experiments 1 and 2: [NO] is constant and [Cl2] is doubled; the rate also doubles (2.52 to 5.04 x 10^-3), so the reaction is first order in Cl2. Compare experiments 1 and 3: [Cl2] is constant and [NO] is doubled; the rate becomes about four times (2.52 to 10.05 x 10^-3), so the reaction is second order in NO. Therefore the rate law is Rate = K [NO]^2 [Cl2]. The overall order = 2 + 1 = 3.

# Chapter 10: Solution & Colloids

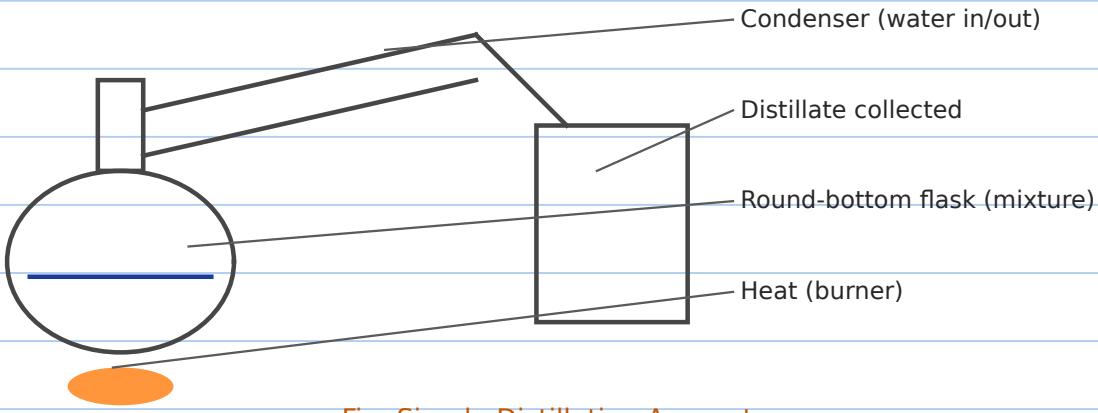



Fig: Simple Distillation Apparatus

## Multiple Choice Questions: 1. Choose the correct answer

(i) Only one pair of liquid in the following set does not obey Raoult law, identify it:

(a) Methanol and Ethanol

(b) Benzene and toluene

(c) n-Hexane and n-heptane

✓ (d) Ethanol and Acetone

Reason: Ethanol and acetone form a non-ideal solution with interactions different from the pure liquids, so this pair does not obey Raoult's law; the other pairs are nearly ideal.

Question: (ii) Identify the incorrect statement about colloidal solution: (a) It shows Tyndall effect (b) Its particles movement is Brownian type (c) Its particle size is less than 1 nm (d) Its physical appearance is translucent

Answer: (c) — Colloidal particles are 1 nm to 1000 nm in size; particles smaller than 1 nm belong to a true solution, so statement (c) is incorrect.

(iii) Effect of pressure change play significant role in the solubility of:

(a) Solid into liquid

(b) Liquid into liquid

✓ (c) Gas into liquid

(d) All of them

Reason: Pressure has a great effect on the solubility of a gas in a liquid (Henry's law), while its effect on solids and liquids dissolving in liquids is negligible.

(iv) Which is not a colligative property:

(a) Lowering in vapours pressure

(b) Elevation in boiling point

(c) Depression in freezing point

✓ (d) Atmospheric pressure

Reason: Colligative properties depend on the number of solute particles: lowering of vapour pressure, elevation of boiling point, depression of freezing point and osmotic pressure; atmospheric pressure is not one of them.

(v) According to Raoult law the relative lowering of vapour pressure is equal to:

(a) Mole fraction of solvent

✓ (b) Mole fraction of solute

(c) Molality

(d) Molarity

Reason: Raoult's law states that the relative lowering of vapour pressure of a solution of a non-volatile solute is equal to the mole fraction of the solute.

(vi) The sum of mole fractions of components of a solution is equal to:

(a) 0.0

✓ (b) 1.0

(c) 10

(d) 100

Reason: Mole fraction is a fraction of the total number of moles, so the sum of the mole fractions of all components of a solution is always 1.0.

(vii) How many moles of NaOH are present in a 2dm3 of 1 molar aqueous solution of it.

(a) 0.5 mole

(b) 1 mole

(c) 1.5 mole

✓ (d) 2 mole

Reason: Moles = molarity x volume = 1 mol/dm3 x 2 dm3 = 2 moles of NaOH.

(viii) A colloidal solution of liquid into liquid is known as:

(a) Gel

(b) Foam

(c) Sol

✓ (d) Emulsion

Reason: A colloid of a liquid dispersed in another liquid is called an emulsion, e.g. milk; a gel is liquid in solid and a sol is solid in liquid.

(ix) An example of completely immiscible liquid pair is:

(a) Benzene to toluene

(b) Water and phenol

(c) Water and Benzene

(d) Water and methanol

Reason: Water (polar) and benzene (non-polar) do not mix at all and form two separate layers, so they are a completely immiscible pair.

(x) A 15% W/W KOH solution can be prepared by mixing 15g KOH in:

(a) 15g water

(b) 85g water

(c) 100g water

(d) 115g water

Reason: Percent weight/weight means 15g solute in 100g solution, so the water needed is 100 - 15 = 85g.

## Short Questions

Question: 1. Name various units of concentration and explain mole fraction.

Answer: The concentration of a solution is expressed in several units: percentage composition (weight/weight, weight/volume, volume/volume), molarity (moles of solute per dm3 of solution), molality (moles of solute per kg of solvent) and mole fraction. Mole fraction is the ratio of the number of moles of a component to the total number of moles of all the components in the solution. For a component A, XA = nA/(nA + nB). It has no units, each mole fraction lies between 0 and 1, and the sum of the mole fractions of all components is equal to 1.

Question: 2. Define Molarity and molality. Which of these depends on temperature?

Answer: Molarity is the number of moles of solute dissolved in 1 dm3 (litre) of solution, denoted by M. Molality is the number of moles of solute

dissolved in 1 kg of solvent, denoted by m. Molarity depends on temperature because the volume of the solution expands or contracts with temperature, so the number of moles per dm3 changes. Molality does not depend on temperature because it is based on the mass of the solvent, which does not change with temperature. For this reason molality is preferred in work involving colligative properties.

Question: 3. Explain on particles bases how the vapours pressure of solution is lowered by adding non volatile solute.

Answer: In a pure solvent, the whole surface consists of solvent particles which escape into the vapour, producing its vapour pressure. When a non-volatile solute is dissolved, some surface positions are occupied by solute particles, which cannot escape into the vapour. Hence fewer solvent particles can leave the surface per unit time, and the rate of evaporation decreases. The vapour pressure of the solution therefore becomes lower than that of the pure solvent. This lowering depends on the fraction of solute particles, i.e. on the mole fraction of the solute (Raoult's law).

Question: 4. What are miscible and immiscible liquids? Why n-hexane (petrol) is immiscible in water?

Answer: Miscible liquids are liquids which mix completely in all proportions to form a homogeneous solution, e.g. water and ethanol. Immiscible liquids do not mix and separate into two layers, e.g. oil and water.

n-Hexane (petrol) is a non-polar liquid whose molecules are held by weak London forces, whereas water is polar and strongly hydrogen bonded. The water molecules attract each other much more strongly than they attract hexane molecules, so hexane cannot break into the water structure; following the rule "like dissolves like", the two liquids remain immiscible.

Question: 5. Solubility of oxygen in water increases with pressure but solubility of glucose in water has negligible effect of pressure why?

Answer: Oxygen is a gas dissolved in a liquid; when the pressure of the gas above the water is increased, more gas molecules strike the surface and are forced into solution (Henry's law), so its solubility increases appreciably with pressure. Glucose is a solid solute; solids and liquids are almost incompressible, so changing the external pressure does not change the closeness of its particles or its dissolving equilibrium to any noticeable extent. Therefore the solubility of glucose in water has a negligible effect of pressure. Pressure mainly affects the solubility of gases, not of solids.

## Descriptive Questions

Question: 1. State Raoult's law and derive its mathematical expression.

Answer: Raoult's law states that the vapour pressure of a solution of a non-volatile solute is directly proportional to the mole fraction of the solvent in the solution. Consider a solution with n1 moles of solvent and n2 moles of solute; the mole fraction of solvent is X1 = n1/(n1 + n2). If P0 is the vapour pressure of the pure solvent and P the vapour pressure of the solution, then by Raoult's law P = P0 X1 = P0 n1/(n1 + n2). Rearranging, the relative lowering of vapour pressure is (P0 - P)/P0 = n2/(n1 + n2) = X2, the mole fraction of the solute. This expression is used to find the molar mass of the solute from the measured lowering of vapour pressure.

Question: 2. What does an Ideal solution means? Give four characteristics to distinguish between ideal and non ideal solution.

Answer: An ideal solution is one which perfectly obeys Raoult's law at all temperatures and concentrations. Four distinguishing characteristics are: (1) In an ideal solution no heat is absorbed or evolved on mixing (DH

mixing = 0), while in a non-ideal solution heat is absorbed or evolved. (2) There is no change in volume on mixing an ideal solution (DV = 0), while a non-ideal solution may expand or contract. (3) In an ideal solution the forces between unlike molecules are the same as between like molecules; in a non-ideal solution they are different. (4) Examples: benzene + toluene and n-hexane + n-heptane are ideal, while ethanol + water is non-ideal and shows a deviation from Raoult's law.

Question: 3. Define osmosis and osmotic pressure. Give four daily life examples of osmosis.

Answer: Osmosis is the spontaneous flow of solvent particles from a dilute solution (higher solvent concentration) to a concentrated solution through a semipermeable membrane. Osmotic pressure is the hydrostatic pressure which must be applied to the solution to stop this flow of solvent across the membrane; it is a colligative property. Daily life examples are: absorption of water by plant roots from the soil; shrinking of raw mangoes and grapes in strong salt or sugar solution; swelling of dried raisins placed in water; and the use of salt to preserve pickles and meat by drawing water out of micro-organisms. Reverse osmosis, used in R.O. plants, is based on applying pressure greater than the osmotic pressure.

Question: 4. What are colligative properties? Why does the boiling point of a liquid get raised when a non volatile solute is added?

Answer: Colligative properties are physical properties of a dilute solution which depend only on the number of solute particles (molecules or ions) present and not on their nature. The main colligative properties are lowering of vapour pressure, elevation of boiling point, depression of freezing point and osmotic pressure. When a non-volatile solute is added, it lowers the vapour pressure of the solvent because some surface

positions are taken by non-volatile solute particles. A liquid boils when its vapour pressure becomes equal to the atmospheric pressure; since the vapour pressure is now lower, the solution must be heated to a higher temperature to reach that pressure. Hence the boiling point is raised (elevated).

Question: 5. Differentiate among true solution, colloidal solution and suspension on the bases of. (i) Particle size (ii) Visibility

Answer: (i) Particle size: in a true solution the particles are the smallest, less than 1 nm (ions or small molecules); in a colloidal solution the particle size is between 1 nm and 1000 nm; in a suspension the particles are larger than 1000 nm. (ii) Visibility: the particles of a true solution cannot be seen even with a powerful microscope; colloidal particles also cannot be seen with an ordinary microscope but can be detected with an ultramicroscope or by the Tyndall effect; the particles of a suspension can be seen with the naked eye or an ordinary microscope and they settle down on standing. A true solution is homogeneous and translucent effects are absent, a colloid is translucent and shows the Tyndall effect, while a suspension is opaque and heterogeneous.

Numerical Questions

Question: 1. Automotive antifreeze is a 60% (w/w) aqueous solution of ethylene glycol ($C_2H_6O_2$). Determine (a) molality of solution (b) mole fraction of ethylene glycol in the solution. [Ans: molality = 24.19mol/kg, X = 0.3]

Answer: Take 100g of solution: it contains 60g ethylene glycol and 40g water. Molar mass of $C_2H_6O_2$ = 2(12) + 6(1) + 2(16) = 62 g/mol, so moles of glycol = 60/62 = 0.968 mol. (a) Molality = moles of solute / mass of solvent in kg = 0.968/0.040 = 24.19 mol/kg. (b) Moles of water = 40/18 =

2.22 mol. Mole fraction of glycol X = 0.968/(0.968 + 2.22) = 0.3.

Question: 2. A solution is prepared by mixing 46g ethanol (C2H5OH) and 180g water. Calculate the mole fraction of both components. [Ans: X2 = 0.090, X1 = 0.909]

Answer: Molar mass of ethanol = 46 g/mol, so moles of ethanol = 46/46 = 1 mol. Moles of water = 180/18 = 10 mol. Total moles = 11. Mole fraction of ethanol X2 = 1/11 = 0.090, and mole fraction of water X1 = 10/11 = 0.909. The sum of the two mole fractions is 1, as required.

# Chapter 11: Thermochemistry

## Multiple Choice Questions: 1. Choose the correct answer

(i) Least entropy found in which of the following state of water:

(a) Liquid water at 4 degree C

(b) Liquid water at 25 degree C

(c) Steam at 100 degree C

✓ (d) Ice at 0 degree C

Reason: Entropy (disorder) is least in the most ordered state; ice is a solid with particles fixed in a lattice, so it has the least entropy of the given states of water.

(ii) The energy corresponds to the given thermochemical process is labeled as: Li+(g) + Cl-(g) -> LiCl(s)

(a) Ionization energy

✓ (b) Lattice energy

(c) Enthalpy of combustion

(d) Enthalpy of atomization

Reason: The formation of one mole of an ionic solid from its gaseous ions releases the lattice energy, so this process represents lattice energy.

(iii) Which of the following change is not an endothermic reaction:

(a) Decomposition of lime

(b) Cracking of alkanes

✓ (c) Combustion of butane

(d) Photosynthesis

Reason: Combustion of butane gives out heat, so it is exothermic;

decomposition of lime, cracking of alkanes and photosynthesis all absorb heat and are endothermic.

(iv) Heat transfer can not be feasible across the boundry of a:

(a) Open system

(b) Close system

(c) Isolated system

(d) Thermo permeable system

Reason: An isolated system exchanges neither matter nor energy (heat) with its surroundings, so heat transfer across its boundary is not possible.

(v) Which of the following enthalpy change is always negative:

(a) Enthalpy of formation

(b) Enthalpy of decomposition

(c) Enthalpy of combustion

(d) Enthalpy of reaction

Reason: Combustion is always an exothermic process in which heat is evolved, so the enthalpy of combustion is always negative.

(vi) Standard enthalpy of formation of all of the following elements at 25 degree C and 1 atm pressure are zero except:

(c) O2

(d) N2

Reason: By convention the standard enthalpy of formation of an element in its most stable form is zero; graphite is the stable form of carbon, not diamond, so C(diamond) has a non-zero value.

(vii) In the equation of First law of thermodynamics (DE = q + w), the property(s) which depends upon initial and final state is (are):

(a) DE

(b) q

(c) W

(d) Both q and W

Reason: Internal energy change DE is a state function and depends only on the initial and final states, whereas q and w are path functions.

(viii) Which statement is incorrect:

(a) For constant pressure process, DH = DE + PDV

(b) For constant volume process, DE = q

✓ (c) For exothermic reactions, DH > 0

Reason: For exothermic reactions heat is evolved and DH is negative (DH < 0), so the statement DH > 0 for exothermic reactions is incorrect.

(ix) In a thermochemical process, no work is done if the system is kept at:

(a) Constant temperature

(b) Constant pressure

✓ (c) Constant volume

(d) Constant mass

Reason: Pressure-volume work W = -P DV; at constant volume DV = 0, so no work is done.

(x) Volume is a:

✓ (a) State function

(b) Path function

(c) Intensive properties

(d) Colligative property

✓

Reason: As given in the book's answer key; volume depends only on the state of the system (it is also an extensive property), not on the path followed.

## Short Questions

Question: 1. State precisely the meaning of each of the following terms. (i) System and surrounding (ii) State and state function (iii) Internal energy and Enthalpy

Answer: (i) The system is the specific part of the universe under experimental study, and everything outside it is the surrounding; the two are separated by a boundary. (ii) The state of a system is its condition described by measurable properties like temperature, pressure and volume; a state function is a property whose value depends only on the initial and final states and not on the path, e.g. internal energy and enthalpy. (iii) Internal energy E is the sum of all kinetic and potential energies of the particles of a system. Enthalpy H is the total heat content of a system at constant pressure, defined as H = E + PV.

Question: 2. What is meant by Internal energy change (DE) and Enthalpy change (DH)? Under what conditions are DE and DH equal?

Answer: Internal energy change DE is the difference in internal energy between the final and initial states of a system; at constant volume it is equal to the heat absorbed, DE = qv. Enthalpy change DH is the heat change at constant pressure, DH = qp, and is related by DH = DE + PDV (or DH = DE + Dn(g) RT). DE and DH are equal when there is no change in volume, i.e. when the reaction is carried out in a closed vessel at constant volume, or when there is no change in the number of gas moles (Dn(g) = 0). They are also nearly equal for reactions involving only solids and liquids, where DV is negligible.

Question: 3. How can you define standard enthalpy of formation and standard enthalpy of reaction.

Answer: The standard enthalpy of formation (DHf degree) is the heat evolved or absorbed when one mole of a substance is produced from its constituent elements in their standard states, i.e. at 25 degree C (298K) and 1 atm pressure; by convention it is zero for elements in their most stable form. The standard enthalpy of reaction (DH degree) is the enthalpy change of a chemical reaction when all reactants and products are in their standard states. It is calculated as DH degree = sum of DHf degree of products minus sum of DHf degree of reactants. These values allow the heat change of a reaction to be found without doing the experiment.

## Descriptive Questions

Question: 1. State and explain First law of Thermodynamics. Derive pressure-volume work of a system.

Answer: The first law of thermodynamics states that energy can neither be created nor destroyed; the total energy of a system and its surroundings remains constant (it is the law of conservation of energy). Mathematically, DE = q + w, where DE is the change in internal energy, q the heat absorbed by the system and w the work done on the system. For pressure-volume work, suppose a gas in a cylinder expands by DV against a constant external pressure P. The work done by the gas = force x distance = P x DV, so the work done on the system is w = -P DV. The negative sign shows that in expansion (DV positive) work is done by the system and its internal energy decreases, while in compression work is done on the system.

Question: 2. Discuss the applications of the First law of thermodynamics at constant pressure and constant volume.

Answer: At constant pressure, no condition is imposed on volume, and the heat absorbed is used partly to increase internal energy and partly to do expansion work; this heat change is the enthalpy change, qp = DH = DE + PDV. Most laboratory and industrial reactions occur at constant (atmospheric) pressure, so enthalpy changes measured in an open apparatus apply the first law in this form. At constant volume, DV = 0, so no pressure-volume work is done (w = 0) and all the heat absorbed appears as internal energy change, qv = DE. This is the basis of the bomb calorimeter, in which the heat of combustion of a fuel or food is measured at constant volume and DE is found directly.

Question: 3. State and explain Hess's Law of enthalpy summation. Discuss its applications.

Answer: Hess's law states that if a reaction is carried out through several steps, the total enthalpy change for the overall reaction is equal to the sum of the enthalpy changes of the individual steps, whatever path is followed. It is a consequence of the first law of thermodynamics and of enthalpy being a state function. Its applications are: calculating enthalpies of formation of compounds which cannot be measured directly (e.g. methane); determining the enthalpy changes of slow or incomplete reactions; constructing Born-Haber cycles to find the lattice energy of ionic compounds; and calculating the calorific values of fuels and foods. In cyclic form it is expressed as the sum of DH around a complete cycle being zero.

Question: 4. Explain Exothermic and Endothermic reactions with the help of the energy diagram.

Answer: In an exothermic reaction heat is evolved to the surroundings, the products have less enthalpy than the reactants, and DH is negative;

burning of fuel and neutralization are examples. In an endothermic reaction heat is absorbed from the surroundings, the products have more enthalpy than the reactants, and DH is positive; decomposition of CaCO3 and photosynthesis are examples. On an energy diagram, enthalpy is plotted against the progress of the reaction, with a peak for the activated complex in between. For an exothermic reaction the product level lies below the reactant level, while for an endothermic reaction the product level lies above it; the vertical difference between the two levels is DH in each case.

## Numerical Questions

Question: 1. A thermochemical process is carried out at constant pressure of 8.52atm. If it absorbs 15.4KJ energy from the surrounding due to which an expansion in the volume of 4.7dm3 is occurred. Calculate its change in internal energy. [Ans: 11.34KJ]

Answer: Work done by the system in expansion: w = -P DV = -8.52 atm x 4.7 dm3 = -40.04 dm3.atm. Converting (1 dm3.atm = 0.1013 KJ): w = -40.04 x 0.1013 = -4.06 KJ. Heat absorbed q = +15.4 KJ. By the first law, DE = q + w = 15.4 - 4.06 = 11.34 KJ. So the internal energy of the system increases by 11.34 KJ.

Question: 2. Using the data in table 11.2, calculate the standard enthalpy change for each of the following reactions. 2H2S(g) + 3O2(g) -> 2H2O(l) + 2SO2(g); Fe2O3(s) + 3CO(g) -> 2Fe(s) + 3CO2(g)

Answer: Use DH degree = sum DHf degree (products) - sum DHf degree (reactants), with the standard enthalpies of formation from Table 11.2 of the book (in KJ/mol: H2S(g) = -20.6, H2O(l) = -285.8, SO2(g) = -296.8, Fe2O3(s) = -822.2, CO(g) = -110.5, CO2(g) = -393.5, elements = 0). First reaction: products = 2(-285.8) + 2(-296.8) = -1165.2; reactants = 2(-20.6)

+ 3(0) = -41.2; DH degree = -1165.2 - (-41.2) = -1124 KJ. Second reaction: products = 2(0) + 3(-393.5) = -1180.5; reactants = -822.2 + 3(-110.5) = -1153.7; DH degree = -1180.5 - (-1153.7) = -26.8 KJ. Both reactions are exothermic.

Question: 3. In the manufacturing of HNO3 by the Ostwald process, one of the most important exothermic reactions is the oxidation of ammonia. 4NH3(g) + 5O2(g) -> 4NO(g) + 6H2O(l) Determine standard heat of reaction (DH degree) from the following given data. DHf degree of NH3(g) = -46.19 KJ/mol, DHf degree of NO(g) = 90.25 KJ/mol, DHf degree of H2O(l) = -285.8 KJ/mol [Ans: -1169.04 KJ/mol]

Answer: DH degree = [4 DHf(NO) + 6 DHf(H2O)] - [4 DHf(NH3) + 5 DHf(O2)]. Products: 4(90.25) + 6(-285.8) = 361 - 1714.8 = -1353.8 KJ. Reactants: 4(-46.19) + 5(0) = -184.76 KJ. DH degree = -1353.8 - (-184.76) = -1169.04 KJ for the reaction as written. The negative sign confirms that the oxidation of ammonia is strongly exothermic.

Question: 4. Iso octane (C8H18) is an efficient fuel with a high octane rating. The combustion of C8H18 in an internal combustion engine is represented in the following thermochemical equation. Find its standard heat of combustion. C8H18(l) + 12.5 O2(g) -> 8CO2(g) + 9H2O(l) (DHc degree = ?) Given that DHf degree of CO2 = -393.5 KJ/mol, DHf degree of H2O = -285.8 KJ/mol, DHf degree of C8H18 = -223.8 KJ/mol [Ans: -5496.4 KJ/mol]

Answer: DHc degree = [8 DHf(CO2) + 9 DHf(H2O)] - [DHf(C8H18) + 12.5 DHf(O2)]. Products: 8(-393.5) + 9(-285.8) = -3148 - 2572.2 = -5720.2 KJ. Reactants: -223.8 + 0 = -223.8 KJ. DHc degree = -5720.2 - (-223.8) = -5496.4 KJ/mol. So burning one mole of iso-octane releases 5496.4 KJ of heat, which is why it is an efficient fuel.

Question: 5. Glycerol (C3H8O3) is a well known organic compound due to its versatile uses. Calculate the standard enthalpy of formation of Glycerol from the data given below. 3C(s) + 4H2(g) + (3/2)O2(g) -> C3H8O3(l) (DHf degree = ?) C(s) + O2(g) -> CO2(g) (DH degree = -393.5 KJ/mol) H2(g) + (1/2)O2(g) -> H2O(l) (DH degree = -285.8 KJ/mol) C3H8O3(l) + (7/2)O2(g) -> 3CO2(g) + 4H2O(l) (DH degree = -1654.1 KJ/mol) [Ans: -669.6 KJ/mol]

Answer: Apply Hess's law. Combine 3 x (formation of CO2) + 4 x (formation of H2O) and subtract the combustion of glycerol: 3C + 3O2 -> 3CO2 gives 3(-393.5) = -1180.5 KJ; 4H2 + 2O2 -> 4H2O gives 4(-285.8) = -1143.2 KJ; reversing the combustion, 3CO2 + 4H2O -> C3H8O3 + (7/2)O2 gives +1654.1 KJ. Adding all steps, the CO2, H2O and extra O2 cancel, leaving 3C + 4H2 + (3/2)O2 -> C3H8O3. DHf degree = -1180.5 - 1143.2 + 1654.1 = -669.6 KJ/mol.

Question: 6. Draw a fully labeled Born Haber cycle for Rubidium chloride (RbCl) and determine the lattice energy by using the following values. (all in kJ/mol). I.P of Rb = 403, Electron affinity of Cl = -349, Bond energy of Cl2 = 242 KJ/mol, Sublimation energy of Rb = 86.5 KJ/mol, Heat of formation of RbCl(s) = -430.5 KJ/mol [Ans: -692 KJ/mol]

Answer: In the Born-Haber cycle, RbCl(s) is formed from its elements by two routes; by Hess's law both routes have the same total enthalpy. Route steps from the elements: sublimation of Rb = +86.5; ionization of Rb = +403; dissociation of half a mole of Cl2 = (1/2)(242) = +121; electron affinity of Cl = -349; lattice energy = U (to be found). Sum of steps = enthalpy of formation: 86.5 + 403 + 121 - 349 + U = -430.5. So 261.5 + U = -430.5, giving U = -692 KJ/mol. The negative sign shows that 692 KJ/mol is released when one mole of RbCl crystal forms from gaseous Rb+ and Cl- ions. (As instructed, the cycle is described in words rather than drawn; its steps are sublimation, ionization, dissociation, electron affinity and lattice

formation.)

# Chapter 12: Electrochemistry

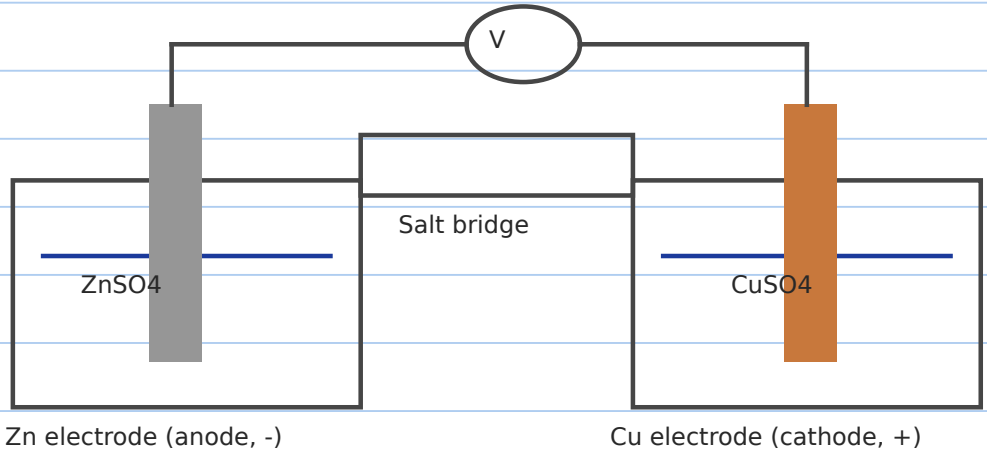



Fig: A Galvanic (Voltaic) Cell — Daniell Cell

## Multiple Choice Questions: 1. Choose the correct answer

(i) The outer body of dry cell serves as anode, it is made up of:

(a) Copper

✓ (b) Zinc

(c) Lead

(d) Iron

Reason: In a dry cell the outer zinc container itself acts as the anode, where zinc is oxidized to Zn+2 ions.

(ii) The conduction of electricity through an electrolytic solution is due to the flow of:

(a) Electrons

✓ (b) Ions

(c) Atoms

(d) Molecules

Reason: In an electrolytic solution the charge is carried by the movement of positive and negative ions towards the electrodes, not

by electrons.

(iii) During electrolysis, the reaction that takes place at anode is:

✓ (a) Oxidation

(b) Reduction

(c) Hydrolysis

(d) Simultaneous oxidation and reduction

Reason: The anode is the electrode where oxidation (loss of electrons) always takes place, in both electrolytic and galvanic cells.

Question: (iv) This statement is not correct for lead storage battery: (a) It can be recharged (b) It is a primary battery (c) Anode is made up of lead (d) Cathode is made up of lead oxide

Answer: (b) — The lead storage battery is a secondary (rechargeable) battery, not a primary battery, so statement (b) is not correct.

(v) In Zn-SHE voltaic cell, the half reaction occurs at anode is:

(a) Zn+2 + 2e- -> Zn

✓ (b) Zn -> Zn+2 + 2e-

(c) 2H+ + 2e- -> H2

(d) H2 -> 2H+ + 2e-

Reason: Zinc is the anode in this cell and is oxidized: Zn -> Zn+2 + 2e-; the electrons flow to the hydrogen electrode (cathode).

(vi) The strongest oxidizing agent in the electro chemical series is:

(a) Li

(b) H2

(c) Cu

✓ (d) F2

Reason: Fluorine lies at the bottom of the electrochemical series with

the highest positive reduction potential, so F2 is the strongest oxidizing agent (Li is the strongest reducing agent).

(vii) Galvanized rode of iron is coated with:

(a) Nickel

✓ (b) Zinc

(c) Chromium

(d) Carbon

Reason: Galvanizing is the coating of iron with a thin layer of zinc, which protects the iron from corrosion by acting as a sacrificial metal.

(viii) Fuel cell is a typical Galvanic cell which is based on the reaction between:

(a) Nitrogen and oxygen

✓ (b) Hydrogen and oxygen

(c) Methane and oxygen

(d) Hydrogen and zinc

Reason: A fuel cell generates electricity from the reaction 2H2(g) + O2(g) -> 2H2O(l), with a continuous supply of hydrogen and oxygen.

(ix) Oxidation number of Cr in Na2Cr2O7 is:

(a) +3

✓ (b) +6

(c) +8

(d) +12

Reason: Let the oxidation number of Cr be x: 2(+1) + 2x + 7(-2) = 0, so 2x = 12 and x = +6.

(x) Which of the following half cell reaction show oxidation:

(a) Fe+3 -> Fe+2

(b) Cl2 -> 2Cl-

(c) SO4 -2 -> SO4 -3

(d) Zn -> Zn+2

Reason: Zn -> Zn+2 + 2e- involves loss of electrons, which is oxidation; the others are reductions (gain of electrons).

## Short Questions

Question: 1. Define the following: Primary and secondary cell; Oxidizing agent and reducing agent; Oxidation and reduction; Oxidation number; Redox reactions

Answer: A primary cell is a galvanic cell which cannot be recharged and is discarded once its chemicals are consumed, e.g. dry cell; a secondary cell can be recharged by passing current in the reverse direction, e.g. lead storage battery. An oxidizing agent oxidizes another substance and itself gets reduced, while a reducing agent reduces another substance and itself gets oxidized. Oxidation is the loss of electrons (increase in oxidation number) and reduction is the gain of electrons (decrease in oxidation number). Oxidation number is the apparent charge an atom has in a compound according to set rules. Redox reactions are reactions involving transfer of electrons, in which oxidation and reduction take place simultaneously.

Question: 2. What is meant by oxidation number? Determine the oxidation number in following. (i) Cr in H2CrO4 (ii) S in K2S4O6 (iii) Fe in Fe3O4 (iv) Cl in HClO4

Answer: Oxidation number (oxidation state) is the charge, real or apparent, which an atom of an element has in a compound or ion, calculated by assuming that shared electrons belong completely to the more electronegative atom. (i) H2CrO4: 2(+1) + x + 4(-2) = 0, so x = +6

for Cr. (ii) $K_2S_4O_6$: 2(+1) + 4x + 6(-2) = 0, so 4x = 10 and x = +2.5 (average) for S. (iii) $Fe_3O_4$: 3x + 4(-2) = 0, so x = +8/3 (average) for Fe. (iv) $HClO_4$: +1 + x + 4(-2) = 0, so x = +7 for Cl.

Question: 3. Define the term standard electrode potential and hydrogen electrode.

Answer: Standard electrode potential (E degree) is the potential difference created between a metal electrode and a one molar solution of its ions at 25 degree C and 1 atmospheric pressure, measured against the standard hydrogen electrode. The standard hydrogen electrode (SHE) is the reference electrode, assigned a potential of 0.0 volt. It consists of a platinum foil coated with platinum black, immersed in a 1M acid solution, with pure hydrogen gas at 1 atm pressure bubbling over it at 25 degree C. All other electrode potentials are measured relative to it, and half-cell reactions are written as reductions by convention.

Question: 4. In the process of electroplating the item to be plated is made cathode, give reason?

Answer: Electroplating deposits a metal from its ions in solution onto the surface of an article by electrolysis. Metal ions are positively charged cations, and they are attracted to the negatively charged electrode, the cathode, where they gain electrons and are deposited as a layer of metal. Therefore the article to be plated must be made the cathode, so that the coating metal is reduced and deposited on it. The anode is usually made of the pure coating metal, which dissolves to keep the concentration of ions in the bath constant. This is the principle used in tin plating, chromium plating and galvanizing-type coatings.

Question: 5. What is corrosion? What causes it to form? What can be done

to prevent its formation?

Answer: Corrosion is an electrochemical process in which atoms on the surface of a metal are oxidized (converted into harmful oxides) due to the action of the surrounding medium, such as moist air; the common example is the rusting of iron, 4Fe + 3O2 -> 2Fe2O3. It is caused by the metal surface acting as tiny anode and cathode areas in the presence of water containing dissolved O2 and CO2, which serves as an electrolyte, so electrons flow and the metal is eaten away. It can be prevented by painting or coating with grease or oil, by galvanizing (zinc coating) or electroplating with tin or chromium, and by alloying the metal, e.g. making stainless steel.

Question: 6. How can you define an electrochemical series? Give its properties.

Answer: The electrochemical series (ECS) is an arrangement of elements (electrodes) in order of their increasing standard electrode (reduction) potential values, with hydrogen in the middle as reference. Its properties are: elements above hydrogen have negative E degree values and are strong reducing agents, the strongest being Li at the top; elements below hydrogen have positive values and their ions are stronger oxidizing agents, F2 being the strongest. A metal higher in the series can displace a metal lower in the series from its salt solution. The series is used to predict the products of electrolysis, the feasibility of a redox reaction and the EMF of a galvanic cell.

## Descriptive Questions

Question: 1. (a) What is meant by electrode potential and standard Hydrogen electrode (SHE). (b) Describe how the electrode potential of copper is determined.

Answer: (a) Electrode potential is the potential difference set up between a metal electrode and the solution of its own ions when the metal is dipped into that solution; it arises because metal atoms tend to lose electrons (oxidation potential) or ions tend to gain them (reduction potential). The standard hydrogen electrode (SHE) is the reference electrode with an assigned potential of 0.0 V; it consists of platinum foil in 1M H+ solution with H2 gas at 1 atm and 25 degree C. (b) To determine the electrode potential of copper, a Cu rod is dipped in 1M CuSO4 solution to form a half cell, which is connected through a salt bridge to a SHE and through wires to a voltmeter. The measured EMF of the cell Pt/H2/H+ // Cu+2/Cu is +0.34 V, and since the SHE potential is zero, the standard electrode potential of copper is E degree = +0.34 V, the positive sign showing copper is below hydrogen in the electrochemical series.

Question: 2. Sketch a copper-zinc Galvanic cell, write the cell reaction and show the direction of electron flow.

Answer: The cell consists of a zinc rod dipped in ZnSO4 solution (anode half cell) and a copper rod dipped in CuSO4 solution (cathode half cell), connected externally by a wire through a voltmeter and internally by a salt bridge which allows ion flow but prevents mixing of the solutions. At the anode oxidation occurs: Zn -> Zn+2 + 2e-. At the cathode reduction occurs: Cu+2 + 2e- -> Cu. The overall cell reaction is Zn(s) + CuSO4(aq) -> ZnSO4(aq) + Cu(s), with a standard EMF of 1.10 V. Electrons flow through the external circuit from the zinc anode (negative terminal) to the copper cathode (positive terminal), while current flows in the opposite direction. (As instructed, the sketch is described in words: two beakers joined by an inverted U-tube salt bridge, Zn electrode on the left losing mass and Cu electrode on the right gaining mass.)

Question: 3. What is the difference between a primary cell and secondary cell sketch a diagram of dry cell and explain its working.

Answer: A primary cell cannot be recharged; once its chemicals are consumed it stops working and is thrown away, e.g. the dry cell. A secondary cell can be recharged by passing electric current in the reverse direction, so it can be used again and again, e.g. the lead storage battery. The dry cell has an outer zinc container which serves as the anode, a central graphite rod as the cathode, and a paste of MnO2, NH4Cl, ZnCl2 and carbon powder around it acting as the electrolyte. At the anode Zn is oxidized to Zn+2 ions and releases electrons; at the cathode MnO2 is reduced while NH4+ ions take electrons. The electrons flow through the external circuit and light a bulb or run an appliance. Its EMF is about 1.5 V, and it leaks and runs down with time because the zinc container is gradually consumed.

Question: 4. Balance the following equations by ion electron method. (i) Cr2O7 -2 + I2 -> Cr+3 + IO3- (acidic medium) (ii) MnO4- + SO3 -2 -> Mn+2 + SO4 -2 (basic medium) (iii) MnO4- + Cl- -> Mn+2 + Cl2 (acidic medium) (iv) Cl2 + OH- -> Cl- + ClO3- + H2O (basic medium) (v) MnO4- + C2O4 -2 + H+ -> Mn+2 + CO2 + H2O (acidic medium) (vi) Fe+2 + Cr2O7 -2 + H+ -> Fe+3 + Cr+3 + H2O (acidic medium)

Answer: By the ion-electron method each equation is split into oxidation and reduction half-reactions, each half is balanced for atoms and charge using H+, OH- and H2O as the medium requires, the electrons are equalized, and the halves are added. (i) Oxidation: I2 + 6H2O -> 2IO3- + 12H+ + 10e-; reduction: Cr2O7 2- + 14H+ + 6e- -> 2Cr3+ + 7H2O; equalizing electrons (LCM 30) and adding gives: 5Cr2O7 2- + 3I2 + 34H+ -> 10Cr3+ + 6IO3- + 17H2O. (ii) In basic medium permanganate is reduced to MnO2, and the balanced equation is: 2MnO4- + 3SO3 2- + H2O

-> 2MnO2 + 3SO4 2- + 2OH- (the book prints Mn+2 as the manganese product, but Mn+2 is not stable in basic medium). (iii) 2MnO4- + 10Cl- + 16H+ -> 2Mn2+ + 5Cl2 + 8H2O. (iv) 3Cl2 + 6OH- -> 5Cl- + ClO3- + 3H2O. (v) 2MnO4- + 5C2O4 2- + 16H+ -> 2Mn2+ + 10CO2 + 8H2O. (vi) 6Fe2+ + Cr2O7 2- + 14H+ -> 6Fe3+ + 2Cr3+ + 7H2O. Every equation above has been checked to balance in both atoms and charge.